

Cross-vendor heterogeneity as a guard against epistemic homogenization in AI-assisted scientific synthesis

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Abstract

AI-assisted synthesis is becoming consequential infrastructure for scientific research, and a new failure mode emerges when a single AI model is the sole synthesizer of a long-lived knowledge corpus: the corpus accumulates that model’s blind spots and biases over time, with each individual edit plausible and internally consistent but the cumulative drift invisible without an outside reference. We call this drift *epistemic homogenization* and distinguish it from per-output hallucination. We argue that the right granularity of defense is **cross-vendor heterogeneity**, routing successive synthesis passes through frontier models trained by different companies, with different training corpora and different RLHF procedures. The pattern has a *defensive* value (catching errors and framing biases aligned with the dominant vendor’s prior) and a *generative* value (surfacing connections the dominant vendor would not propose unprompted regardless of how rigorous its self-review pipeline is). We describe a production deployment as a two-layer pattern: an interactive hypothesis-development layer (author plus Anthropic Claude Opus) and an automated sweep daemon (DeepSeek V4-Pro at propagation and synthesis with Google Gemini 2.5 Pro as automatic Pass 2 fallback; OpenAI GPT-5.5 at review). The strongest evidence in the paper is reflexive: while drafting this manuscript, the cross-vendor review pass caught a magnitude error in the internal wiki documentation that the paper was citing as evidence (§5.0); the corrected figure appears in §5.3 and the catch is verifiable as a literal diff against the manuscript’s revision history. Six additional operational case studies (§5.1–§5.6) extend the same pattern across approximately three weeks of operation on a ~95-page personal-research wiki on gout and engineered koji as a therapeutic-enzyme chassis, including one boundary-case limitation (§5.6: corpus-level contamination inherited identically across vendors from training data, defended by independent verification-of-verification chains rather than by cross-vendor heterogeneity). A pilot empirical study of inter-vendor agreement (§6) reports mean pairwise cross-vendor agreement of ~82% across 8 prompts × 4 vendors and a within-vendor replicate study at ~91%, partitioning the ~18% cross-vendor disagreement into roughly half within-vendor temperature noise and roughly half real cross-vendor heterogeneity (approximately 8 percentage points of real prior-distribution difference under the pilot’s manual claim-coding scheme). The disagreement is concentrated in the task types the architecture is designed to exploit. The contribution is a methodology case study with N=1 operational deployment plus pilot empirical evidence; controlled ablations comparing cross-vendor against alternative review architectures remain future work.

§1 Introduction

AI-assisted synthesis is becoming infrastructure for how scientific research is conducted. Research groups across biotechnology, chemistry, physics, and the life sciences are deploying frontier language models to read primary literature, propose connections across papers, draft hypotheses, and maintain working knowledge graphs that span hundreds or thousands of source documents. The pattern is no longer experimental; it is operational infrastructure.

This shift introduces a new class of failure that does not exist in traditional human-only research. When a single AI model is the sole synthesizer of a long-lived knowledge corpus, the corpus accumulates that model’s blind spots

and biases over time. Each individual synthesis output is reviewed and edited, but the review itself runs through the same model, propagating the same prior. New entries cross-reference older entries that were already shaped by the same model’s framings. After hundreds of edits, the corpus no longer represents the underlying literature; it represents the literature *as that model sees it*. We call this drift **epistemic homogenization**.

Epistemic homogenization is structurally different from per-output hallucination. A hallucinated answer is a single wrong claim that any careful reader can catch. Homogenization is a property of the corpus as a whole, consistent biases compounding across hundreds of internally-coherent edits, no single output obviously wrong, the cumulative drift invisible without an outside reference. A reader querying the homogenized corpus receives confident, internally consistent answers that drift from reality in a direction the reader cannot easily perceive. The corpus’s defenses (peer review of individual edits; cross-referencing; evidence-tagging) operate within the same prior that produced the drift in the first place.

The natural response is to introduce diversity into the synthesis pipeline, run multiple models in review, alternate which model writes which sections, ensemble the outputs. The existing literature on multi-agent AI systems (debate, self-refine, jury-of-LLMs, Constitutional AI) develops these patterns substantially. This work operates predominantly *within* a single vendor: multiple instances of the same company’s model, or multiple sizes of the same company’s model family. Within-vendor heterogeneity is primarily multi-instance, not multi-perspective at the prior-distribution level. It does not target blind spots that are characteristic of the vendor’s training pipeline as a whole.

This paper argues that **cross-vendor heterogeneity**, models trained by different companies, on different (though overlapping) data corpora, with different reinforcement-learning-from-human-feedback (RLHF) procedures and different alignment objectives, is a level at which substantial prior-distribution diversity reliably appears. A blind spot in one vendor’s training pipeline is less likely to appear in the same form in another vendor’s, though correlated blind spots from shared upstream data exist and are discussed as a limitation in §7. The cross-vendor pattern exploits the partial independence that does exist, in the same way classical ensemble methods exploit cross-architecture independence, but at a vendor abstraction level that the existing multi-agent literature has not explicitly developed.

The contribution of this paper is threefold. First, we develop the cross-vendor heterogeneity guard as an explicit architectural pattern, formalize the failure class it protects against (epistemic homogenization), and distinguish it from the within-vendor patterns in the existing multi-agent literature (§4). Second, we distinguish two complementary values of the pattern, a **safety-net** value (defensive: catching errors the within-vendor pipeline misses) and a **search-amplifier** value (generative: surfacing connections the within-vendor pipeline does not propose), and argue that the search-amplifier value is what makes cross-vendor add a capability not supplied by verification alone. Third, we describe an operational deployment as a two-layer pattern: an *interactive hypothesis-development layer* (author plus Anthropic Claude Opus) where new wiki content originates, and an automated *sweep daemon* (DeepSeek V4-Pro at Pass 1 and Pass 2 with Google Gemini 2.5 Pro fallback, OpenAI GPT-5.5 at Pass 3) that propagates and reviews wiki edits. The full system has covered three-to-four distinct vendors across approximately three weeks of production use on a ~95-page personal-research wiki on gout, hyperuricemia, and NLRP3-driven inflammation, with engineered koji (*A. oryzae*) for therapeutic enzyme production as the primary chassis under investigation (and roughly 90% of project focus). We present seven case studies organized around a leading reflexive example (§5.0) plus six from the operational record: four safety-net catches (§5.1–§5.4), one search-amplifier case (§5.5) in which two independent cross-vendor models converged on three platform-level findings the dominant pipeline had missed, and one boundary-case limitation (§5.6) in which corpus-level contamination inherited identically across vendors from training data defeated cross-vendor review and was caught instead by an independent verification-of-verification chain. §5.0 is the empirical core: while drafting this manuscript, the cross-vendor review pass caught a magnitude error in the internal wiki documentation that the paper was citing as evidence; the corrected figure now appears in §5.3 and the catch is verifiable against the manuscript’s revision history. The paper’s posture is methodology case study with N=1 deployment plus a small pilot empirical study (§6) and qualitative catches, not broad empirical proof; we frame the operational record as case-study evidence and queue controlled ablations as future work.

The remainder of the paper is organized as follows. §2 positions the cross-vendor pattern against prior multi-agent AI literature. §3 describes the architecture of the Open Enzyme deployment. §4 develops the heterogeneity-guard rationale and the two-values framing. §5 presents the seven case studies, organized around a leading reflexive example (§5.0) where this manuscript’s own production caught a magnitude error in the wiki documentation it was citing, followed by six operational cases (§5.1–§5.6). §6 reports operational data (cost, latency, failure modes) and a pilot empirical study of inter-vendor agreement. §7 enumerates the architecture’s limitations and the failure classes it does *not* protect against. §8 discusses generalization (how many vendors is enough, scaling to non-text modalities, implications for autonomous-research systems, the human in the loop). §9 concludes. A Methods

Appendix documents the cross-vendor process used to produce this manuscript itself, the paper is drafted using the methodology it describes, including a logged self-catch where the primary drafter confabulated a fact that primary-source verification would have caught.

§2 Related work

The cross-vendor heterogeneity guard described in this paper sits in a literature already rich with multi-agent and self-reflective patterns for improving language-model outputs. This section positions the contribution against five methodology clusters in that literature, plus a sixth body of work on the underlying corpus-collapse phenomenon that motivates the guard.

These clusters share a structural property the paper’s contribution does not. They operate either *within* a single model-vendor’s training pipeline, or at an abstraction level orthogonal to vendor identity altogether. The pattern this paper develops, heterogeneity at the level of training-distribution prior, achieved by routing successive synthesis passes through different vendors’ frontier models, is, to our knowledge, the first explicit treatment of the cross-vendor regime as an architectural commitment for AI-assisted scientific synthesis.

2.1 Multi-Agent Debate and Consensus Approaches

Du et al. [1] introduced multi-agent debate as a mechanism for improving the factuality and reasoning of language-model outputs: multiple instances, typically of the same model, though Du et al. also explored mixed-model configurations such as ChatGPT and Bard in debate, propose and critique one another’s answers across several rounds before converging on a final response. The work demonstrated gains on reasoning benchmarks and reductions in characteristic hallucination patterns, and the framing has been adopted and extended by follow-on work on adversarial argumentation and consensus mechanisms.

The structural commitment of multi-agent debate is at the inference-time argumentation layer. The heterogeneity in the system is in how the instances interact, proposing, critiquing, defending, not in their underlying training-distribution priors. When the instances all derive from a single vendor’s model family, the system improves robustness to inference-time stochasticity but does not guard against blind spots that are characteristic of the vendor’s training pipeline as a whole. The cross-vendor pattern in this paper operates at a different abstraction level: heterogeneity is in the prior itself, with the synthesis passes routed across vendors as an architectural commitment rather than as inference-time agent behavior.

2.2 Self-Refinement and Iterative Reflection

Madaan et al. [2] proposed Self-Refine, in which a single language model produces an initial output and then critiques and revises that output in an iterative loop. Shinn et al. [3] proposed Reflexion, which extends the same pattern by maintaining verbal reflective text in an episodic memory buffer so that an agent can learn from prior trials without weight updates. Both patterns are influential and have spawned a substantial body of follow-on work on self-critique loops.

These approaches are, by construction, same-vendor and typically same-model: the critic is the generator. They improve local output quality and per-task accuracy, but they cannot surface blind spots that are characteristic of the model’s training pipeline, the critic shares the generator’s prior. A self-refinement loop catches consistency errors and surface-level mistakes; it cannot catch corpus-level drift toward the model’s prior across many independent outputs maintained in a long-lived knowledge graph. The cross-vendor heterogeneity guard described here is complementary to self-refinement rather than a substitute: self-refinement improves individual outputs; the cross-vendor guard prevents the corpus they accumulate into from converging on any one model’s blind spots.

2.3 LLM-as-Jury and Panel-of-Evaluators

Verga et al. [4] proposed evaluating LLM generations using a panel of diverse models (PoLL), an ensemble of LLM evaluators rather than a single large judge, and showed that smaller-model panels can outperform a single large judge on standard evaluation benchmarks while exhibiting less intra-model bias. The work motivates panel diversity by appeal to the same intuition the present paper formalizes for synthesis: a single judge’s biases propagate into the evaluation signal; a panel of diverse judges reduces that effect.

The PoLL pattern is itself an early example of cross-vendor ensemble in the AI-evaluation literature: the original Verga et al. construction uses Cohere’s command-r, OpenAI’s gpt-3.5-turbo, and Anthropic’s Claude haiku as the panel, three distinct vendors. This makes PoLL the closest antecedent in the existing literature to the cross-vendor heterogeneity guard developed here. The distinction this paper draws is in the *abstraction level of application*. The jury pattern operates on per-output evaluation: each query gets an independent panel review, and there is no temporal accumulation of bias across queries to guard against. The cross-vendor heterogeneity guard operates on *corpus-level synthesis*: sequential passes through different vendors maintaining a long-lived knowledge graph across hundreds of edits. Both inherit the same structural intuition, vendor diversity reduces propagated bias, but apply it to different operational surfaces.

2.4 Constitutional AI and AI-Feedback Alignment

The broader AI-feedback alignment family includes Ouyang et al. [5] InstructGPT, which established RLHF (reinforcement learning from human feedback) as a frontier-scale technique for shaping language-model behavior; Bai et al. [6] Constitutional AI, which extended the pattern by training a harmless assistant through self-supervision against an explicit list of principles without per-output human labels; and Lee et al. [7] RLAIIF, which uses an off-the-shelf LLM rather than a human annotator to generate the preferences that drive reward-model training. The family spans multiple vendors (OpenAI, Anthropic, Google) but the techniques are all *training-time* interventions, they shape a single model’s parameters through AI or human feedback at training time, not at inference time.

The cross-vendor heterogeneity guard described here operates at synthesis time, using different-vendor frontier models as the sources of independent critique on a corpus that none of them owns or controls. The two families are adjacent but address disjoint failure modes: alignment techniques affect a single model’s prior; the cross-vendor guard prevents that prior from dominating a corpus the model contributes to over time.

2.5 Automated AI-Scientist Systems

A class of recent end-to-end automated-research systems aspires to compress the full scientific lifecycle, hypothesis generation, experiment design, analysis, and manuscript drafting, into a single agentic pipeline. Lu et al. [8] introduced Sakana’s AI Scientist, an early system aimed at end-to-end open-ended scientific discovery in machine learning, with AI-generated papers accepted at workshop-level venues. Yamada et al. [9] extended the system with agentic tree search and further demonstrated AI-generated papers passing workshop-level peer review under a different review threshold. Song et al. [10] introduced PaperOrchestra, a system that automates manuscript drafting through five communicating agents (outline, plotting, literature review, section writing, content refinement), with citation verification against Semantic Scholar. Go et al. [16] introduced LiRA, a more focused multi-agent framework targeting the literature-review subtask specifically, outlining, subsection writing, editing, and reviewing agents producing review articles that outperform single-agent and tree-search baselines on standard survey benchmarks.

Two structural features unite these systems and motivate the relevance of cross-vendor heterogeneity to autonomous-research deployments specifically. First, each has been primarily demonstrated within a single vendor’s model family in published evaluations, Claude or GPT for AI Scientist, Gemini-based for PaperOrchestra. The architectures do not technically require single-vendor construction, but the published evaluations are single-vendor pipelines. Second, they iterate AI synthesis as a critical step where each builds on the prior step’s AI output, so any blind spot in the system’s prior compounds across loop iterations before a human can intervene. The cross-vendor heterogeneity guard is one structural defense against this compounding. None of the systems above currently deploys it as an architectural commitment, though PaperOrchestra’s Semantic-Scholar citation-verification step is a partial defense against one failure mode (fabricated citations). The pattern this paper develops can be adopted by such systems with modest engineering effort and provides a defense the within-vendor architectures do not.

(Disclosure: this manuscript’s §2 was drafted using a community implementation of the PaperOrchestra writing pipeline, see the Methods Appendix for the production-process record.)

2.6 Model Collapse, Knowledge Collapse, and Output Homogenization

The underlying phenomenon the cross-vendor heterogeneity guard defends against sits in a small but rapidly developing literature on AI-content feedback loops. Several adjacent constructs are worth distinguishing.

Model collapse (Shumailov et al., first as a 2023 arXiv preprint “The Curse of Recursion” and subsequently as a 2024 *Nature* publication “AI models collapse when trained on recursively generated data”) describes a *training-time* failure: when a generative model is trained on data produced by previous generations of the same model, the

distribution of generated outputs progressively loses its tails and degenerates over time. The two cited papers are the same research line; the preprint introduced the phenomenon and the *Nature* publication established the canonical reference.

Knowledge collapse (Peterson [13], “AI and the Problem of Knowledge Collapse,” subsequently published in *AI & Society*) describes a *societal*-level failure: widespread reliance on recursive AI systems can paradoxically narrow public understanding because LLMs systematically generate outputs toward the center of their training distribution, and that center becomes the reference for users discounting alternative sources. Peterson’s model shows that a 20% discount on AI-generated content can move public beliefs $2.3\times$ further from the truth than no discount, with recursive interdependence between AI systems dramatically accelerating the effect.

Epistemic diversity / knowledge-collapse measurement (Wright et al. [14], “Epistemic Diversity and Knowledge Collapse in Large Language Models”) operationalizes the construct by measuring variation in real-world claims across 27 LLMs on 155 topics. Their finding: all tested models produce less epistemically diverse output than a basic web search; model size has a negative effect on diversity, while retrieval-augmented generation has a positive effect. This is empirical evidence that LLM output homogenization is real and measurable, and that the gap from web-search baseline widens with scale.

Output homogenization as a distinct failure mode (Rudko and Bonab [15], “ChatGPT is incredible (at being average),” *Ethics and Information Technology*) argues that output homogenization is structurally distinct from hallucination and bias, and proposes the distinction between “what-BS” (falsehoods, factual inconsistencies) and “how-BS” (homogenization of output structure). This paper explicitly frames homogenization as under-studied relative to hallucination.

Epistemic homogenization as the present paper uses the term is the *corpus-level* analogue of these phenomena: a knowledge graph maintained by AI synthesizers drifts toward the synthesizer’s prior over many edits, in the same direction Peterson and Rudko/Bonab describe at the societal and per-output levels, and in a manner consistent with Wright et al. [14]’s empirical finding that scale alone reduces rather than increases epistemic diversity. Our contribution is to develop a *runtime architectural pattern* (cross-vendor heterogeneity at synthesis time) that targets the corpus-level surface specifically. Wright et al. [14]’s empirical methodology suggests an obvious future direction: applying their epistemic-diversity metric to the Open Enzyme corpus over time as a quantitative validation of the cross-vendor pattern. That ablation is queued as future work; the present paper offers the operational case study, not the quantitative metric.

The cross-vendor heterogeneity guard does not address training-time model collapse directly, nor does it address societal knowledge collapse; it addresses the corpus-level analogue specifically. The four are complementary defenses for related failure modes in the AI-content-feedback-loop literature. We treat epistemic homogenization as a distinct construct, while acknowledging the shared structural intuition: any system that consumes its own outputs as essential input over time develops a degradation that requires an outside reference to detect.

§3 Architecture

The deployment described in this paper is a **two-layer cross-vendor pattern**: an interactive hypothesis-development layer where the author drives synthesis with an Anthropic Claude Opus session, and an automated sweep daemon that fires on every wiki edit and routes propagation, synthesis, and review across distinct vendor models. The heterogeneity guard applies across both layers; vendor counts and case-study catches in §5 are reported for the full pattern, not the daemon alone. Figure 1 shows the architecture; the rest of §3 walks each layer in turn.

Figure 1. The two-layer architecture: an interactive hypothesis-development layer (author + Anthropic Claude Opus) feeding the wiki corpus, and an automated sweep daemon (DeepSeek Pass 1 and 2, Google Gemini Pass 2 fallback, OpenAI GPT-5.5 Pass 3) that propagates and reviews each wiki edit. An episodic independent-vendor peer-review pass sits off the per-edit critical path. Figure source: `figures/figure1_architecture.py`.

Layer 1 - The interactive hypothesis-development layer

The first layer is the author’s working session with Anthropic Claude Opus inside Claude Code. The pattern at this layer is the lab analogue of AI-assisted interactive coding: leading with curiosity, asking why and why not dozens of times, picking the threads worth pulling, naming the sub-claims a downstream experiment will need to test, running computational experiments before any wet-lab spend. This is where new wiki content originates - new mechanistic

Figure 1. The two-layer cross-vendor cycle

Layer 1 originates new wiki content. Layer 2 propagates and reviews it automatically. The human triages the queue and the cycle restarts.

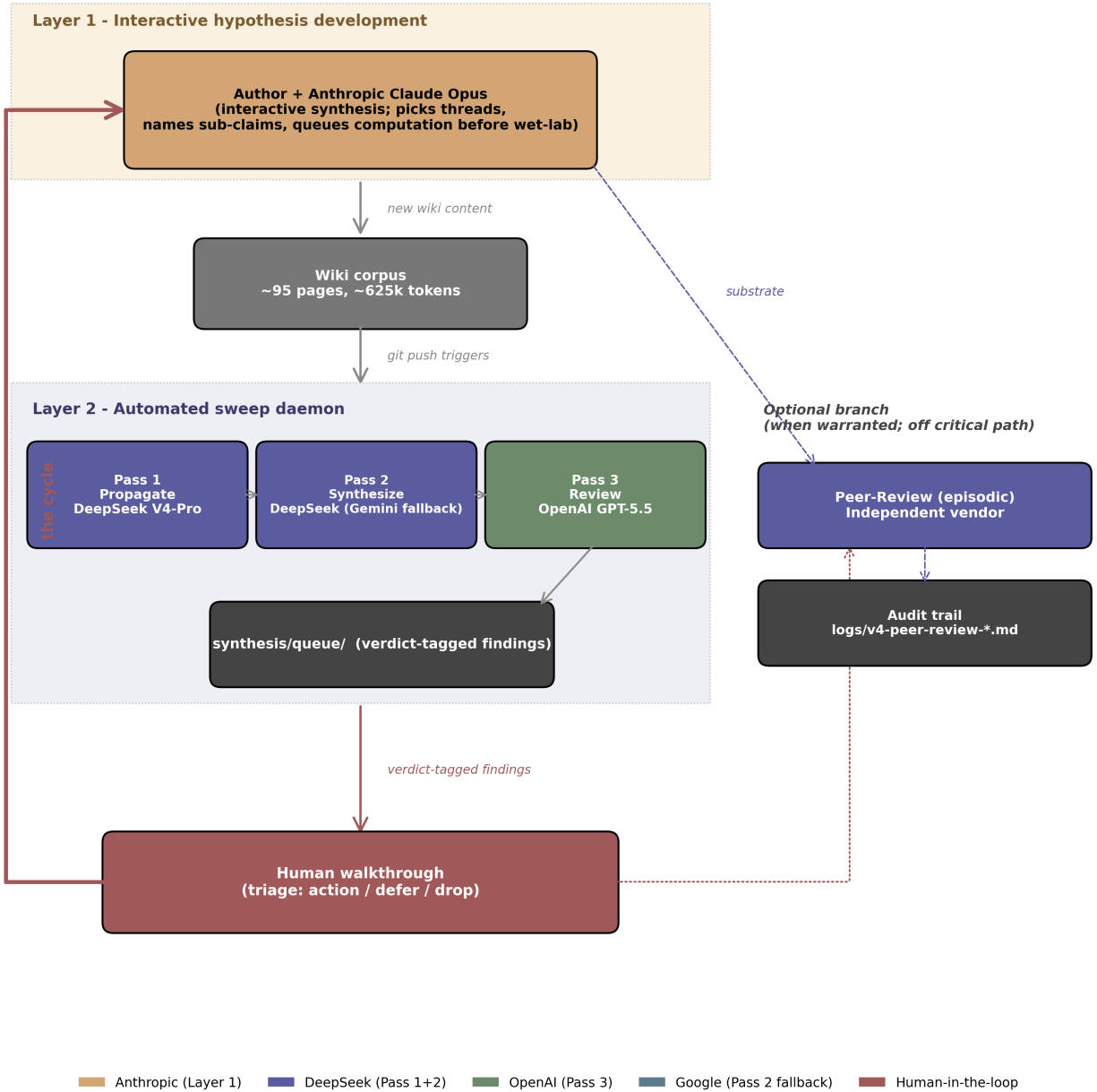


Figure 1: Figure 1. Cross-vendor heterogeneity guard architecture

hypotheses, new numbered computational-experiment briefs, new cross-page connections, new entries in the wiki’s knowledge graph. The wiki state that the automated daemon then propagates and reviews is, in operational terms, the output of this interactive layer.

Layer 1’s vendor contribution to the cross-vendor pattern is **Anthropic Claude Opus**. This is the only point in the system where Anthropic models appear in the per-edit critical path; the automated daemon does not currently route any pass to Anthropic (Claude Opus 4.7 remains available as a Pass 3 alternate, but GPT-5.5 has been the default since 2026-05-08 after a three-way eval at `evals/pass-3-reviewer/`).

Layer 2 - The sweep daemon (three automated passes)

The automated layer fires on every push to `wiki/*.md`. It orchestrates three OpenRouter-routed passes against models from two distinct vendors (DeepSeek + OpenAI), with a Google fallback path on Pass 2:

Pass 1 - Propagate (DeepSeek V4-Pro). Reads the edited trigger file(s); identifies the concepts, compounds, organisms, and mechanisms touched; greps the rest of the corpus for pages that reference those concepts; updates each affected page inline. Includes a dedup discipline so that recurrent propagation does not produce parallel copies of the same content (`scripts/sweep-prompt-1-propagate.md`). Every new claim is tagged with an evidence level (Clinical Trial / Animal Model / In Vitro / Mechanistic Extrapolation) and inline provenance (`((source: <filename>))`). Invoked from `.github/workflows/wiki-sweep.yml` line 185 with `--model deepseek/deepseek-v4-pro`.

Pass 2 - Synthesize (DeepSeek V4-Pro primary, Google Gemini 2.5 Pro automatic fallback). Reads the full corpus (currently ~625k tokens) and emits cross-document synthesis: new connections, contradictions, open questions, proposed experiments (`scripts/sweep-prompt-2-synthesize.md`). The two models are configured via OpenRouter’s `models` array; Gemini fires when the DeepSeek call fails (context overflow, transient API error). The DeepSeek-primary/Gemini-fallback split is operational: `synthesize.py`’s history block records that Pass 2 was on Gemini-primary 2026-04-25 → 2026-05-05 and was re-routed to DeepSeek-primary 2026-05-05 after a head-to-head evaluation in which DeepSeek produced higher-quality findings at lower per-call cost. Gemini was retained as the automatic fallback because DeepSeek’s OpenRouter route was operationally flaky during the eval window (transient API errors); the primary assignment is empirically validated, the fallback exists for reliability. Each emitted finding ends with a `{{PEER-REVIEW}}` marker so Pass 3 knows where to insert a review.

Pass 3 - Review (OpenAI GPT-5.5; Anthropic Claude Opus 4.7 alternate). Critiques each Pass 2 finding with a fixed verdict vocabulary (*Confirmed*, *Confirmed-prioritize*, *Partial*, *Push-back*, *Rejected*). The reviewer prompt is model-tuned: `scripts/sweep-prompt-3-review-gpt55.md` for GPT-5.5, `scripts/sweep-prompt-3-review.md` for Claude. GPT-5.5 became the default on 2026-05-08 after a three-way eval. Pass 3 has read-only tool access (`read_file`, `list_files`, `grep`) for spot-checking primary sources. A deterministic Python emitter then writes one file per reviewed finding into `synthesis/queue/`, where the author closes items by appending a closure annotation and moving the file to `synthesis/done/`.

Vendor count across both layers

The full system covers **three to four distinct vendors**: - **Anthropic** (Layer 1: interactive hypothesis-development) - **DeepSeek** (Layer 2: Pass 1 propagate + Pass 2 synthesize primary) - **OpenAI** (Layer 2: Pass 3 review) - **Google** (Layer 2: Pass 2 synthesize fallback, fires when DeepSeek call fails)

When Pass 2 routes to DeepSeek successfully (the common case), the daemon’s per-edit critical path covers two vendors (DeepSeek + OpenAI) and the full system covers three (adding Anthropic from Layer 1). When Pass 2 fallback fires, Google enters and the full system covers four. The cross-vendor episodic peer-review pass adds a further independent-vendor verification surface (e.g., DeepSeek V4-Pro reviewing an Anthropic-drafted substrate, as in §5.4’s seminal 2026-04-25 catch) outside the per-edit critical path.

This is the architectural commitment the paper formalizes. The specific vendor-to-layer assignments are eval-driven and rotate: the project runs head-to-head evaluations against representative sweep outputs (see the 2026-05-08 Pass 3 switch to GPT-5.5 at `evals/pass-3-reviewer/`, and the 2026-05-05 Pass 2 switch to DeepSeek described above) and slots the winning model into the relevant pass. The pattern - heterogeneity across vendor training pipelines, applied at multiple layers of an AI-assisted research workflow - is the load-bearing claim; the current assignment is the current eval winner, expected to roll over as new models ship and as corpus size constraints change.

Episodic peer-review pass

A separate, formalized **independent peer-review pass** is documented in `wiki/open-source-platform.md` §“Multi-model synthesis as guard against epistemic homogenization” and used episodically rather than on every sweep. An independent vendor’s model receives the substrate (the wiki corpus at a given commit) and produces a parallel synthesis plus a differential analysis against the corpus’s current state. The vendor is deliberately chosen to be distinct from the dominant vendor that drove the substrate, even when that means routing to a vendor not currently in the per-edit critical path - for example, when the substrate is heavily Layer-1-driven (Anthropic), the peer-review pass routes to DeepSeek or Google. The seminal instance (DeepSeek V4-Pro reviewing an Anthropic-driven substrate on 2026-04-25, `logs/v4-peer-review-2026-04-25-deepseek.md`) is the case study in §5.4. Peer-review passes are run when a major architectural change lands, when a class of synthesis output is suspect, or when the cost of a missed connection is judged high enough to justify the additional vendor pass.

Inter-pass artifact handoff and infrastructure resilience

Two failure classes surfaced after initial deployment, both forms of *information loss between passes* rather than within-pass errors. Both motivated changes to the architecture that are orthogonal to model selection.

Information loss inside the architecture. Pass 2 received only the original trigger file list, not Pass 1’s propagated files, so Pass 2’s attention was biased toward triggers even though it read the full corpus. Pass 3 had no way to verify Pass 2 citations of files outside the trigger set. Both failures motivated explicit artifact handoff:

- Pass 1 emits a `propagated_files` list (wiki pages modified beyond the original triggers); Pass 2 receives both `trigger_files` and `propagated_files`, and is instructed that new cross-document connections are most likely to emerge from the union of the two sets.
- Pass 2 emits a `cited_files` manifest listing every wiki page referenced in any finding; Pass 3 receives the synthesis log plus both file lists inlined as a “warm cache,” with read-only tools for fetches the cache misses.

Information loss to CI infrastructure. On 2026-04-28, Pass 1 completed and pushed; Pass 2 ran to completion and generated a synthesis log at \$0.73 cost; then `git push` failed with `! [rejected] main -> main (fetch first)` because concurrent commits had landed on `main` between Pass 1’s push and Pass 2’s push. Pass 3 never ran. The synthesis artifact existed only in the ephemeral GitHub Actions runner and was discarded on teardown. A separate incident the same week saw an OpenRouter HTTP 503 terminate the shell wrapper before any retry could land. Recovery motivated `git pull --rebase origin main` before every inter-pass push (commit `05f8c74`), `retry-with-backoff` (5s / 15s / 45s) wrapping every OpenRouter call (commit `b5dd4e0`), and a `failed-sweep-<sha>.md` ledger uploaded as a workflow artifact so failure traces survive runner teardown. §6.2 documents both incidents with run numbers and architectural responses.

The pattern: every pass must declare what it produced for the next pass to consume, and the inter-pass plumbing must survive concurrent activity on the underlying branch. Both disciplines apply at the infrastructure layer, independent of which models the passes use. Cross-vendor heterogeneity guards against vendor-specific model behaviors; it does not guard against race conditions or transient API outages, and those need their own architectural treatment.

Operational hardening

Beyond the three core passes, the architecture includes five operational components (`scripts/SWEEP-ARCHITECTURE.md`) that emerged from observed failures during early deployment:

1. **Workflow hardening**, `rebase-before-push` to handle concurrent commits to `main`; `retry-with-backoff` (5s / 15s / 45s) on OpenRouter HTTP 5xx; a `failed-sweep-<sha>.md` ledger uploaded as a workflow artifact so failure traces survive runner teardown.
2. **State registry**, `logs/sweep-state.json`, an atomic-write source of truth for “what has been swept and what hasn’t.” Replaces a brittle regex over commit messages.
3. **Hooks for enforcement**, filesystem `pre-commit` hook plus a Claude Code `PostToolUse` hook on `Bash`, both checking the same invariants on `[skip-wiki-sweep]` marker usage and `sweep-N-` commit-message prefixes.
4. **Skills (user-invocable operations)**, `/sweep-status`, `/sweep-catchup`, `/sweep-validate`, slash commands that read the registry and act on it.
5. **Catch-up cron watchdog**, daily safety net that re-fires the workflow if the registry shows pending paths older than 24 hours, with a 3-consecutive-failure brake to prevent runaway cost.

Together, the three operational passes plus five hardening components produce a self-healing pipeline that recovers from concurrent-push races, transient API outages, and operator typos without manual intervention.

§4 The heterogeneity-guard rationale

Defining the failure mode: epistemic homogenization

A knowledge corpus maintained by a single AI synthesizer over time accumulates that synthesizer’s blind spots and biases. New entries are written, reviewed, and cross-referenced through the same model, so consistent errors in the model’s prior, preferences for certain framings, recurrent misreadings of specific terminology, characteristic numerical hallucinations, propagate uncorrected. The corpus stops being a faithful synthesis of the underlying literature and starts being a synthesis of the literature *as that model sees it*. We call this drift **epistemic homogenization** of the corpus.

Epistemic homogenization is structurally distinct from per-output hallucination. A hallucinated answer is a single wrong claim; it may be caught by any reader who looks at it. Homogenization is a property of the *corpus as a whole*: consistent biases compounding over hundreds of edits, each individually plausible, with no single output obviously wrong. A reader querying the homogenized corpus receives confident, internally consistent answers that drift from the underlying reality in a direction the reader cannot easily perceive without an outside reference.

Two values of the heterogeneity guard: safety net and search amplifier

The cross-vendor heterogeneity guard has two complementary values. They are distinct enough that they justify the architecture on independent grounds. Cross-vendor heterogeneity is both a safety net and a search amplifier.

The **safety-net (defensive)** value is what the term “guard” most directly suggests: cross-vendor review catches errors, hallucinations, and framing biases that align with the dominant vendor’s prior and that within-vendor review systematically misses. §5.1–§5.4 below are case studies of the safety-net value, within-vendor cascade, upstream-of-subagent contamination, external-tool reliability failure, and methodological risk invisible from within. These are catches that prevent the corpus from drifting; they reduce the variance of corpus quality over time.

The **search-amplifier (generative)** value is less obvious but more interesting: cross-vendor review *proposes* connections, mechanisms, and reframings that the dominant vendor would not have surfaced unprompted, regardless of how rigorous its self-review pipeline is. The space of useful syntheses any one model can surface is bounded by what its training-distribution prior makes salient. A connection between two corpus pages that aligns with DeepSeek’s mechanistic priors but not with Claude’s may simply never be proposed by Claude even on careful re-reading. Not because Claude is wrong, but because the connection is not in the part of synthesis-space Claude’s prior renders accessible. §5.5 is a case study of the search-amplifier value, showing two different cross-vendor models converging on three platform-level findings the within-vendor pipeline had not made.

The safety net bounds corpus drift; the search amplifier extends corpus reach. Both contribute to the heterogeneity guard’s overall benefit, and which one matters more in any given deployment depends on which is more expensive to lose: an error that propagates uncaught, or a connection that never gets made. For consequential scientific corpora the answer is usually “both,” which is why this paper develops cross-vendor heterogeneity as an architectural commitment rather than as a defensive technique alone.

Why cross-vendor specifically

The natural response to epistemic homogenization is to introduce diversity into the synthesis pipeline, run multiple models in review, or alternate which model writes which sections. But the granularity of that diversity matters. Running GPT-4o and GPT-4 in alternation provides limited defense against OpenAI-specific blind spots; both share substantial overlap in training-distribution prior, alignment procedure, and characteristic failure modes, even though they differ in architecture and training cutoff. Running Sonnet and Opus in alternation similarly provides limited defense against Anthropic-specific blind spots. Within-vendor diversity is non-zero, but it does not target the failure mode this paper is centrally concerned with: drift toward a particular training pipeline’s prior. Within-vendor heterogeneity is primarily multi-instance, not multi-perspective at the prior-distribution level.

Cross-vendor heterogeneity, models trained by different companies, on different (though overlapping) data corpora, with different reinforcement-learning-from-human-feedback (RLHF) pipelines and different alignment objectives, is a

level at which substantial prior-distribution diversity reliably appears. Anthropic-trained models have characteristic verbal patterns and characteristic failure modes that differ from OpenAI-trained models, which differ from Google-trained models, which differ from DeepSeek-trained models. A blind spot in one vendor’s training pipeline is less likely to appear in the same form in another vendor’s pipeline, though correlated blind spots from shared upstream data (e.g., overlapping Common Crawl ingestion) do exist and are discussed as a limitation in §7. The cross-vendor pattern exploits the partial independence that does exist.

The pattern is not novel as a *concept*, it is a special case of ensemble methods, where the heterogeneity of the ensemble members is what produces the variance-reduction benefit. What is novel here is the application of ensemble logic at the *vendor* level for AI-assisted scientific synthesis, where the relevant variance is in the synthesizer’s training-distribution prior, not in inference-time stochasticity. Existing multi-agent literature (debate, self-refine, jury-of-LLMs) operates predominantly within a single vendor; this paper argues that for long-lived scientific corpora where corpus quality has downstream consequences, within-vendor heterogeneity alone is insufficient and the cross-vendor regime should be developed as an explicit architectural commitment.

Capability profile and exposed subjectivity

Two further dimensions of cross-vendor heterogeneity matter for the architectural argument, distinct from the bias-distribution defense developed above.

Capability profile. Vendors differ in native capabilities, not just in training-distribution priors. Anthropic exposes prompt caching via explicit `cache_control` breakpoints; OpenAI exposes structured outputs with JSON-schema conformance; Google’s Gemini ships with natively fluent multilingual support that other vendors approximate via post-training. The Open Enzyme deployment exploits these differences. Pass 3 caches the inlined Pass 2 evidence under Anthropic’s `cache_control` when routed to Opus 4.7 (91% cache-hit rate on representative sweeps; `evals/pass-3-reviewer/2026-05-07-abc8de9-comparison.md`); the GPT-5.5 Pass 3 prompt uses outcome-first ordering and decision-rule phrasing per OpenAI’s published guidance (commit 5d92079); the multilingual ingestion discipline documented in `CLAUDE.md` §“Global-multilingual research by default” assumes the substrate model has native fluency in source-language scientific literature. Vendor selection is consequential for cost and depth of operation, not only for bias coverage.

Exposed subjectivity. Cross-vendor review does not always resolve disagreement; sometimes it exposes it. On 2026-05-07, the same Pass 2 synthesis log (6 findings, 17 trigger files, ~144K tokens inlined evidence) was reviewed in parallel by Anthropic Opus 4.7 and OpenAI GPT-5.5 with prompts tuned to each vendor’s published guidance. Opus tagged 2 of 6 findings as NOVEL, 3 as EXTENSION, 1 as RESTATEMENT. GPT-5.5 tagged 0 NOVEL, 2 EXTENSION, 4 RESTATEMENT (`evals/pass-3-reviewer/2026-05-07-abc8de9-comparison.md`). Both reviewer outputs are internally coherent and defensible by their respective verdict criteria. GPT-5.5’s stricter NOVEL bar (no element of the finding may be named anywhere in the wiki) is more conservative than Opus’s; the disagreement is at the level of verdict criterion, not factual claim. A single-vendor pipeline produces one tag distribution and treats it as the answer. The cross-vendor read produces two, and surfaces the choice of verdict criterion as a thing to consciously calibrate rather than inherit silently.

The self-demonstrating moment

The motivation for the architecture described in this paper did not originate from Claude (Anthropic), the model that drove most of the early Open Enzyme sweep work. It originated from DeepSeek V4-Pro, in a peer-review pass on a Claude Opus 4.7 local-session sweep dated 2026-04-25. DeepSeek’s Connection 7 (a numbered finding in the peer-review log) flagged that the proposal to run cheap full-corpus sweeps on a single model (at the time an attractive cost-optimization) would erode exactly the diversity that makes the corpus trustworthy (`logs/v4-peer-review-2026-04-25-deepseek.md`).

That a model from a different vendor caught a methodological risk Claude had not surfaced unprompted is the cleanest available demonstration of the heterogeneity-guard principle. The principle that later motivated the architecture was demonstrated in advance by the cross-vendor review pass that prefigured it, at the time of Connection 7, the formalized daemon did not yet exist; the catch came from a one-off independent peer-review pass that the formalized architecture is a generalization of. This is the foundational observation the paper extends: cross-vendor review tends to surface methodological risks that within-vendor review systematically misses, because the missed risks are precisely the ones that align with the within-vendor model’s prior.

The self-demonstrating moment is a hybrid of the defensive and generative values: Connection 7 was both a defensive catch (it flagged a homogenization risk that Claude’s pipeline had not surfaced) and a generative one (the framing of “epistemic homogenization” as a distinct construct emerged from DeepSeek’s analysis, not from the substrate it reviewed). The seminal moment did not just catch an error in the substrate, it *named a new concept* that the substrate had not named. The same peer-review run surfaced two further generative findings the substrate had missed (the androgen-urate axis as therapeutic ceiling, and carnosine as androgen-specific counter-agent), which §5.5 develops as the generative-value case study.

§5 Case studies

This section presents seven case studies organized around a single leading example. §5.0 is the load-bearing demonstration: the paper’s own production process catching a documentation error about the very phenomenon it was describing, with the corrected magnitude now in this very subsection. §5.1–§5.4 are four defensive catches from the operational record; §5.5 is a generative case study; §5.6 is a boundary-case limitation showing what the architecture does *not* catch (corpus-level contamination from training data). The defensive cases illustrate failure classes the cross-vendor architecture is designed to catch. The generative case documents the search-amplifier value of cross-vendor review. The boundary case documents what the architecture’s defenses leave to other disciplines (independent verification-of-verification chains). Cases §5.1–§5.6 are drawn from the operational record of 2026-04-25 to 2026-05-08, a 13-day window during which the pipeline ran continuously across approximately 60 wiki edits.

On the biological substrate. Every case study below uses content from the Open Enzyme research wiki: gout pathophysiology, hyperuricemia, uricase enzymology, NLRP3 inflammasome biology, engineered koji (*A. oryzae*) as a therapeutic-enzyme chassis. The biology is workflow substrate: it provides the corpus content over which the cross-vendor pipeline operates and against which catches are measured. Each load-bearing mechanistic claim referenced in a case study is anchored to the primary-source wiki page that carries its evidence-level tagging and citations (`wiki/<page>.md` paths are given inline), and to the audit artifacts that record the catch and correction (`operations/notable-moments.md`, `revisions.md`, git history). The biological evaluation belongs in those wiki pages, not in this paper’s prose: the question this paper asks is whether each catch is *representative* of a failure class an AI-assisted scientific corpus would encounter regardless of biological domain. Readers who wish to interrogate the biology directly are welcome to follow the wiki links and engage the underlying scientific record; readers focused on the AI-methodology argument can take the biological references as substrate without further audit.

§5.0 Leading example, the paper catches an error in its own source material (2026-05-13)

The most demonstrative single piece of evidence this paper offers is reflexive: while drafting this manuscript, the cross-vendor review process caught a magnitude error in the internal wiki documentation that this paper was citing as the empirical basis for §5.3 below. The corrected magnitude appears in §5.3 in its accurate form. This subsection tells the recursive part of the story directly because the audit trail is the contribution.

The original Open Enzyme wiki page `wiki/paperclip-deep-dive.md`, written in the 2026-05-05 operational sweep that motivated §5.3 below, documented the Paperclip MCP `map` operator’s misreport of a Najjari 2022 uricase kinetic parameter. The wiki page’s prose recorded the magnitude of the misreport as “~7,500×” (an arithmetic inversion of the actual ratio: a tool returning 7 μM versus a true value of 52.61 μM is off by a factor of $\sim 7.5\times$, not $\sim 7,500\times$, because the comparison is value-to-value not value-to-fraction). The wrong magnitude entered the corpus on 2026-05-05 and persisted through several downstream sweep cycles without being flagged.

While drafting §5.3 of this manuscript on 2026-05-13, the cross-vendor review pass on the draft compared the manuscript’s prose against the original primary source (Najjari M, et al. *Sci Rep* 12:19913, 2022, PMC9773812). The primary source records the relevant K_m as 52.61 μM (0.05261 mM). The cross-vendor reviewer caught that the magnitude claimed in the source wiki ($\sim 7,500\times$) was internally inconsistent with the actual values reported by both the primary paper (52.61 μM) and the tool under test (7 μM). The correct magnitude is $\sim 7.5\times$. Both the wiki page and §5.3 of this manuscript were corrected on 2026-05-13 (commit `bcf6bae` in the project repository). The Appendix B revision log preserves the audit trail.

The recursive structure of the catch is worth naming explicitly. The wiki page documenting an external tool’s hallucination of a kinetic parameter itself contained a hallucinated description of that hallucination’s magnitude. The within-vendor pipeline that produced both the wiki page and the §5.3 draft did not surface either error; both

were caught only when a different-vendor model reviewed the same content with an explicit primary-source-cross-check brief. This is the failure mode §5.1 (the DAF SCR1-4 disulfide-count cascade) names at the wiki-authoring scope, and the failure mode §5.4 (DeepSeek Connection 7) names at the methodology scope, instantiated again at the **manuscript-production scope**. The pattern repeats at every scope where an AI-assisted corpus accumulates load-bearing numerical claims without primary-source verification at the point of authorship.

What this case study uniquely provides that §5.1, §5.2, §5.3, §5.4, §5.5, and §5.6 do not is *temporal proximity to the reader*. The other case studies are operational catches from a 13-day window 2026-04-25 to 2026-05-08 that the reader must trust the audit trail for. The §5.0 catch happened **during the production of this manuscript** and is verifiable against the manuscript’s own revision history (`papers/cross-vendor-heterogeneity-guard/revisions.md`, Catch 23 in the working log). The same cross-vendor pattern this paper proposes as a discipline is the pattern that caught an error in this paper’s own source material, before the paper was finalized. A reviewer who wants to verify the mechanism can compare the pre-correction draft (git history before commit `bcf6bae`) against the corrected version (this draft) and see the catch as a literal diff.

This case study foregrounds what the reviewer literature has called the most convincing single feature of operational AI-assisted-science discipline: the corpus actively defending itself against its own accumulated errors at the moment of publication, not as a post-hoc audit, and visible to outside reviewers as a verifiable revision trail. The remaining six case studies are drawn from the operational record and demonstrate the same pattern at different scopes. §5.0 is the case study in which the reader can be the verifying vendor.

§5.1 Within-vendor cascade: the DAF SCR1-4 disulfide-count hallucination (2026-05-06)

On 2026-05-05, a Sonnet subagent was tasked with authoring the wiki page `daf-cd55-scr14-truncated-computational.md`, the computational write-up for comp-012, a stalk-truncation analysis of human DAF/CD55 (a complement regulator, UniProt P08174). The subagent’s brief specified the analysis pipeline (AlphaFold-pLDDT-thresholded exposed-site mapping combined with MEROPS protease cleavage prediction) but did not require disulfide-bond modeling, and indeed, the comp-012 pipeline does not compute disulfide bonds. The Limitations section of the analysis explicitly states “Disulfide bonds not modelled.”

The subagent’s output asserted, in four separate places of prose narrative, “3 conserved disulfide bonds per SCR domain → 12 total for SCR1-4.” The number was not computed by the pipeline; it was generated as plausible-sounding context. The error propagated overnight: the next-morning sweep’s Pass 1 incorporated the “12 disulfide” figure into the H05 hypothesis card (`wiki/hypotheses/H05-daf-scr14-cp0-thesis.md`), and downstream synthesis used it as an input to a chaperone-orthogonal stacking calculation, producing a panic over a fabricated total of 29 disulfides across a triple-cassette design (“17+12=29 disulfides, 1.8× Huynh”) (`CLAUDE.md`).

The catch came from the combination of the next-day sweep and a manual walkthrough verification against the canonical UniProt P08174 record. UniProt has exactly 8 DISULFID feature annotations across SCR1-4, the canonical sushi/CCP fold of 2 disulfides per domain (Cys36-Cys81, Cys65-Cys94 [SCR1]; Cys98-Cys145, Cys129-Cys158 [SCR2]; Cys163-Cys204, Cys190-Cys220 [SCR3]; Cys225-Cys267, Cys253-Cys283 [SCR4]), well established in complement biology. The discrepancy was logged as a correction note on the comp-012 page (`wiki/daf-cd55-scr14-truncated-computational.md`).

This catch motivated the addition of a “pre-commit grep-verify gate” to the wiki authoring discipline (`CLAUDE.md`), which requires every load-bearing quantitative claim to be checked against its primary source *before* the commit lands. The sweep is a backstop; the pre-commit gate is where this class of error should die.

Class of failure caught: within-vendor cascade, a single-model hallucination propagated through multiple downstream pages, plausible enough to evade casual review, caught by primary-source cross-checking under combined sweep + manual discipline.

§5.2 Upstream-of-subagent contamination: the comp-018 framing bias (2026-05-08)

On 2026-05-08, a comp-018 subagent was launched to perform a literature scan of upstream complement-cascade modulators. The brief composed for the subagent included a contrived example, originally from a casual user message: “if the compound is in rosemary, I’ll grow rosemary.” The framing was meant to illustrate the kind of finding that would be useful (a specific, growable, dietary modulator) but it landed verbatim in the subagent’s instructions.

The subagent’s output prioritized rosmarinic acid (a polyphenol abundant in rosemary) as a Tier-1 finding. The subagent’s report-back literally said “Brian’s literal ‘if it’s in rosemary I’ll grow rosemary’ framing landed empirically”,

the rhetorical-callback tell that signaled the framing had influenced not just the search but the framing of the result. On review, the evidence supporting Tier-1 placement was thinner than the subagent’s narrative suggested: the IC50 spread across assays was $44\times$ (not the $20\text{--}30\times$ the report acknowledged), and the central mechanism was upstream covalent C3b modification, not direct C5 convertase inhibition.

The catch came from comp-020, a verification rerun specified without the contaminated framing. comp-020’s brief was deliberately stripped of contrived examples and asked the same literature scan as an open question. The rerun found three tied tier-1 candidates instead of one (rosmarinic acid, *Helicteres* benzofuran lignans, luteolin); comp-018 had missed *Helicteres* lignans entirely despite their potency advantage ($4\text{--}20\times$ more potent than rosmarinic acid on matched assay); marine sulfated polysaccharides were underweighted. The discrepancy was logged in `operations/comp-018-vs-comp-020-retrospective.md` (`operations/comp-018-vs-comp-020-retrospective.md`).

This catch motivated the “subagent brief hygiene” discipline now documented in the project’s working instructions (`scripts/SWEEP-ARCHITECTURE.md`). The full statement of the rule is at SWEEP-ARCHITECTURE.md §“Subagent brief hygiene: scope propagates, predictions don’t.” Underlying findings in comp-018 were not contaminated; headline-promotion *was*; coverage breadth was *partially* contaminated.

Class of failure caught: upstream-of-subagent contamination, user-supplied framing biases the subagent’s output before any model-internal reasoning happens, caught only by independent rerun under a cleaner brief. This class is invisible to within-pass review of the contaminated subagent’s own output, because the output is internally coherent.

§5.3 External-tool reliability test: the Paperclip MCP probe (2026-05-05)

The Paperclip MCP (a Model Context Protocol server exposing approximately 11 million full-text scientific papers across PMC, arXiv, and bioRxiv/medRxiv via a unified query interface) was evaluated as a potential augmentation to the sweep daemon, specifically as a “literature delta” surface that would let the pipeline detect when a wiki page had fallen behind recent publications. The candidate integration point was the Paperclip `map` operator, a synthesis primitive that promised structured extraction across multiple papers.

The evaluation took the form of a ground-truth probe: ~ 12 papers from the uricase variant landscape (a domain where Open Enzyme already has hand-curated ground truths for organism, kinetic parameters, and experimental modality). The Paperclip `map` operator was queried for the same fields across these papers, and its outputs were compared against the curated ground truths (`wiki/paperclip-deep-dive.md`).

The probe surfaced three failure classes. (a) Wrong organism identity: papers on *A. flavus* uricase were repeatedly attributed to *A. globiformis*; a separate paper on a porcine-human exon-replacement chimera was attributed to *P. chrysogenum*. (b) Misreported kinetic parameters: for the Najjari 2022 PASylated uricase paper (PMC9773812), the true K_m is $52.61\ \mu\text{M}$ ($= 0.05261\ \text{mM}$); Paperclip’s `map` returned $0.007\ \text{mM}$ ($= 7\ \mu\text{M}$), a ~ 7.5 -fold misreport. The misreport is not a unit-conversion artifact in the tool; both values are presented in stated units. The discrepancy is a hallucinated value at the kinetic-parameter level. The original wiki documentation of this incident initially recorded the misreport magnitude as “ $\sim 7,500\times$ ”, which was itself an arithmetic error; that documentation error was caught by the cross-vendor review of the present manuscript and corrected to the actual $\sim 7.5\times$ figure. See §5.0 for the recursive-catch story; this subsection focuses on the underlying Paperclip-tool probe. (c) Invented wet-lab data for purely computational papers: papers that contained no experimental work were summarized as having produced expression yields and activity measurements.

The decision: do not integrate Paperclip into the sweep pipeline. Wiring the `map` operator into Pass 1 or Pass 2 would inject a structured external hallucination source into a corpus designed for rigor, exactly the failure mode the heterogeneity guard is meant to prevent. Paperclip remains available as a manual research tool used with verification discipline; `map` outputs are never propagated without `grep`-verification.

Class of failure caught: external-tool reliability test before integration. The catch happened because the tool was probed against known-correct ground truths *before* being wired into the pipeline. Without the probe, Paperclip’s plausible-sounding outputs would have entered the corpus through Pass 2 synthesis and contaminated downstream pages.

§5.4 Cross-vendor catches methodological risk: DeepSeek Connection 7 (2026-04-25)

Prior to 2026-04-25, the Open Enzyme sweep work was driven primarily by a single Claude session, Claude Opus 4.7 performing both propagation and synthesis as a local working session, with no formal multi-pass

daemon. The substrate was the wiki at commit 4a40f74. On 2026-04-25, DeepSeek V4-Pro was run as an independent peer-review pass on that substrate: same corpus, parallel synthesis, differential analysis (logs/v4-peer-review-2026-04-25-deepseek.md).

DeepSeek’s Connection 7 in that review observed: “The ‘DeepSeek V4 Assessment’ inadvertently highlights a risk of epistemic homogenization in the sweep process. [...] If a single model (even a frontier one) becomes the sole source of synthesis, the project’s knowledge graph could converge on that model’s blind spots and biases. [...] The value of this corpus is that it can be interrogated by multiple AI systems with different reasoning architectures, creating a form of ‘adversarial collaboration.’”

This was, at the time, a routine peer-review finding among several Connections in the same log. It was not flagged with special urgency. But the observation was consequential: it named a structural risk that the Claude-driven session had not surfaced across previous edits, despite Claude having authored both the synthesis content and the prior self-reviews of that content. The within-vendor pipeline could not see the within-vendor blind spot. A different vendor named it on the first pass.

The architectural response was to formalize the cross-vendor pattern as a daemon: the three operational passes described in §3, deliberately assigned across Anthropic / DeepSeek / OpenAI-or-Anthropic vendors so that no single vendor’s prior could dominate the corpus’s evolution. The peer-review pattern itself was preserved as a separate, episodically-run independent verification mechanism documented in wiki/open-source-platform.md (wiki/open-source-platform.md).

Class of failure caught: methodological risks invisible from within the dominant vendor’s pipeline. This is the highest-value catch class because it is, by construction, the class that within-vendor review is structurally unlikely to surface, the missed risks are precisely the ones that align with the within-vendor model’s prior. It is also the class that justifies the additional cost of the cross-vendor pass: cheaper architectures (within-vendor multi-model, same-model self-critique) are less likely to catch this class, and the operational record reported here is consistent with that pattern, though a controlled ablation comparing cross-vendor against alternative review architectures remains future work.

§5.5 The generative class: cross-vendor as idea discovery (2026-04-25 double-peer-review)

§5.1–§5.4 are catches of the defensive class, errors, contaminations, reliability failures, methodological risks. §5.5 is a catch of the generative class developed in §4: a case where two independent cross-vendor models surfaced *novel connections* that the within-vendor pipeline had not made, regardless of how carefully that pipeline reviewed its own outputs.

On 2026-04-25 the same Claude Opus 4.7 substrate (sweep at commit 4a40f74) that produced the Connection 7 defensive catch in §5.4 was reviewed by two independent peer-review passes: DeepSeek V4-Pro (logs/v4-peer-review-2026-04-25-deepseek.md) and Google Gemini 2.5 Pro (logs/v4-peer-review-2026-04-25-gemini.md). Both peer-review runs concluded with explicit “Missed by Claude” enumerations. **Both vendors independently surfaced the same three findings under that heading.** Convergent independent rediscovery across two vendors with substantially different training-distribution priors is a stronger signal that the catches are real and structural than a single-vendor catch would be.

The three converged findings:

(1) **The androgen-urate axis as a therapeutic ceiling for the gut-lumen sink, not a stratification variable.** Testosterone suppresses ABCG2 expression in the intestinal epithelium; ABCG2 is the transporter the gut-lumen sink strategy depends on to pull urate from blood into the gut lumen. The wiki already contained `androgen-urate-axis.md`, but the within-vendor synthesis had treated it as a patient-stratification note, relevant for dose-sizing in male versus female patients. Both peer-review passes independently reframed it: the male gout demographic, which is the platform’s primary target population, has a *baseline ceiling* on uricase efficacy regardless of dose, because the sink transporter is already pharmacologically downregulated by endogenous androgens. This is a structural constraint on platform feasibility, not a stratification adjustment. DeepSeek’s language was “fundamental, sex-specific constraint on the gut-lumen sink’s maximum efficacy”; Gemini’s was “hard cap on the efficacy of the gut-lumen sink by suppressing the ABCG2 transporter ... reframes the feasibility for the primary male demographic.” Both proposed the same downstream experiment: a Caco-2 ABCG2-transport assay with uricase under androgen-treated versus untreated conditions.

(2) Carnosine as an androgen-specific counter-agent, not a generic NLRP3 synergy. The wiki contained `carnosine.md` (URAT1/GLUT9 downregulation in hyperuricemic rat models) and `androgen-urate-axis.md` (testosterone-driven URAT1 upregulation) on separate pages. The within-vendor synthesis had treated carnosine as a generic dual-pathway candidate (lowers urate, suppresses NLRP3). Both peer-review passes independently connected the two pages: carnosine’s URAT1 downregulation is *specifically tailored* to counteract androgen-driven URAT1 upregulation, making it a uniquely positioned co-expression module for the male gout demographic identified in finding (1). Both passes recommended elevating the carnosine co-expression validation experiment to Phase 1 priority.

(3) Fructose challenge test as an acute n-of-1 efficacy readout. The wiki contained `fructose-connection.md` (KHK pathway, ATP depletion, predictable post-fructose serum urate spike) and `self-experiment-protocol.md` (chronic baseline tracking via serial UA measurements). The within-vendor synthesis treated fructose only as a dietary risk factor. Both peer-review passes independently proposed using the KHK pathway as a controlled acute stress-test for the engineered uricase: standardized 50 g fructose load, serial UA measurements at 0/30/60/90/120 minutes, before and after koji therapy. A blunted post-fructose spike provides direct mechanistic validation of in-gut enzyme activity, in hours rather than weeks. This is a *new protocol*, derived from connecting two existing wiki pages that the substrate had not connected.

Downstream impact (the multiplier on the catch). The three findings did not stay in the peer-review log. Within twelve days of the 2026-04-25 catch, the project absorbed all three: the androgen-urate axis catch drove three computational experiments (comp-016, comp-017, comp-019), one hypothesis card (H-AN-08), and a new dedicated wiki page (`androgen-natural-modulation.md`); the carnosine connection was propagated 2026-05-05 (“synthesis 2026-05-05 item 2/14: carnosine → androgen-URAT1 axis, precision countermeasure for male gout”); and the fructose challenge protocol was incorporated into both `validation-experiments.md` and `self-experiment-protocol.md`. Total cost of the double-peer-review pass that surfaced them was \$0.87 (DeepSeek \$0.21 + Gemini \$0.66).

Class of failure caught: structural under-connection of corpus content. The wiki was internally consistent and contained all the source material; the within-vendor synthesis pipeline had read those pages and had not made the connections that the cross-vendor passes did. No primary-source verification step would have surfaced these findings, they are not errors to be caught against an external reference, they are connections that align with a different training prior than the substrate’s. The pre-commit grep-verify gate (§5.1) is the wrong tool for this class of catch; the cross-vendor pass is the right tool. The §5.5 class is therefore the central argument for why cross-vendor heterogeneity adds a capability not supplied by verification alone: verification catches errors against a fixed reference, but it cannot propose new connections.

On convergent versus divergent generative catches. A natural question is whether the convergence between DeepSeek and Gemini in §5.5 is unusual, or whether different vendors would more typically surface *divergent* generative findings. Both happened in the 2026-04-25 run: the three convergent catches above, plus four other connections DeepSeek surfaced that Gemini did not (BPC-157 gut-healing, bio-AI tooling moat, koji-endgame strain CP3 gap, the Connection 7 homogenization observation itself) and three Gemini surfaced that DeepSeek did not. Convergent catches are high-signal evidence that the substrate vendor missed something structural; divergent catches expand the surface of available syntheses without the convergent-vendor cross-check. Both are valuable; this case study emphasizes the convergent class because it most cleanly demonstrates that the misses were not noise from one reviewer’s prior.

A second instance: human-AI convergence (2026-05-08). The search-amplifier pattern is not confined to AI-AI convergence between cross-vendor review passes. On 2026-05-08 a Gemini-driven Pass 2 synthesis ran on the operational sweep at commit `e842754` and surfaced a finding suggesting the DAF SCR1-4 incident (§5.1 of this paper) should be developed into a public case study on AI-assisted scientific verification. Earlier the same day, before the sweep daemon fired, the project’s author had independently published this incident on a personal blog ([Grounding the AI Scientist hype](#)). The Pass 3 reviewer (GPT-5.5, recently swapped into that slot) pushed back on Gemini’s finding as “already covered”, narrowly true, but scoped to the wiki, because the daemon’s read scope is by design wiki-only (the personal blog repo is sibling-private under the project’s one-way privacy gradient). The synthesizer was right about what to do, the human had done it six hours earlier, the reviewer correctly couldn’t verify the external action. This is the search-amplifier pattern at a different substrate level: a Gemini-based wiki synthesizer running unattended and a human writer drafting a blog post over coffee converged on the same external-comms angle without coordination. The same multi-vendor heterogeneity that catches wrong numbers also catches valuable opportunities, in this instance with a human acting as one of the vendors in the convergent pair. The closed-system property of the daemon (correctly wiki-scoped per the project’s privacy architecture) does not undermine the convergence; it means coordination of action *outside* the daemon’s scope stays a human responsibility, as it should.

§5.6 Corpus-level contamination caught by verification-of-verification (2026-05-07)

The defenses developed in §5.1–§5.5 target failure classes that distribute differently across vendors. A separate class distributes *identically*: contamination inherited from training data. When a claim has been laundered through enough secondary sources that every major frontier model treats it as common knowledge, cross-vendor review provides no defense, because all vendors arrive at the same wrong answer.

The 2026-05-07 sweep on the testosterone-axis adjuvant landscape surfaced an instance with a material biology consequence. A multilingual literature-scan subagent flagged the “37% testosterone elevation” figure cited across *Eurycoma longifolia* (tongkat ali) supplement-marketing pages as suspicious for being too uniform across supposedly independent sources. Verification proceeded in three layers, each a separate subagent run.

Layer 1 (the headline figure). The 37% number traces to Talbott 2013 (PMID 23705671), reporting *salivary* testosterone in a mixed-sex, moderately-stressed cohort, not serum free testosterone in hypogonadal men. Supplement marketing has reproduced the figure for over a decade with that distinction collapsed. The error is at the level of biology: salivary T in non-hypogonadal subjects under moderate stress measures something biologically distinct from serum free T in hypogonadal men.

Layer 2 (the supporting citation). A separate verification subagent, tasked with checking whether the cited papers physically exist, found that the “Shin KH 2024 enclomiphene vs clomiphene” reference appearing on both parent and daughter wiki pages does not exist in PubMed. The actual paper is Saffati et al. 2024 (PMID 39434750). The within-page citation passed casual review because it looked well-formed.

Layer 3 (the mechanism, with material consequence). A third subagent (comp-015 v2) was tasked with verifying the prior wiki claim that eurycomanone, the principal active constituent of tongkat ali, is a xanthine-oxidase inhibitor. Reading the two cited PMIDs’ actual full text, neither paper claims xanthine-oxidase inhibition. The verified mechanism is multi-target purine-handling modulation: URAT1, GLUT9, ABCG2, and NPT1 transporter modulation plus PRPS-suppression. The wiki tag flipped from GOUT-UNFAVORABLE (v1, assumed XO inhibitor: would raise serum urate by blocking catabolism) to GOUT-FAVORABLE (v2, multi-target urate-excretion enhancer; commit c32a623). Supplement marketing has had a gout-favorable compound tagged as gout-unfavorable for a decade.

Why cross-vendor review is unlikely to catch contamination at this layer. All three layers describe contamination at the level of *what well-trained models broadly absorb from secondary literature*. The 37% figure appears in supplement-marketing pages indexed at scale and plausibly distributes across the training corpora of major frontier models. The Shin 2024 citation is plausibly auto-generated by a hallucinating model in some prior secondary source and has propagated. The XO-inhibitor claim for eurycomanone reflects a once-common reading of an early in vitro assay that has been superseded but not retracted from the secondary literature. A second vendor reviewing the wiki page is less likely to flag the verdicts because the contamination is upstream of the vendor. We did not run multi-vendor verification on these three claims as part of this investigation, so the prediction is on theoretical grounds; an empirical multi-vendor check would tighten the claim.

Why verification-of-verification did. The defense that worked is independent of cross-vendor heterogeneity. An independent subagent (running on the same vendor as the original synthesizer, in fact) was given the explicit task of checking the verification subagent’s claims against primary source. Where the first verification had said “Talbott 2013 reports a 37% T increase,” the second opened PMID 23705671 and read the abstract: salivary, mixed-sex, moderate stress. The primary source contradicted the secondary literature, and the independent subagent could see it. Same model, different read. Total cost of the three-layer verification chain: ~\$3-4 in subagent tokens (operations/notable-moments.md 2026-05-07 entry; commits a61f0d9, c32a623).

Class of failure caught: corpus-level contamination. Plausibly distributes broadly across major-vendor training corpora because the laundered claim appears in widely-indexed secondary literature. Defended by independent verification chains targeting primary source rather than by cross-vendor heterogeneity. The discipline applies whenever the substrate model could plausibly be regurgitating a laundered secondary-literature claim. This is common in supplement, over-the-counter, and TCM literatures; it is also common in mainstream medical literature where superseded mechanisms persist in secondary review papers.

§6 Operational data

The Open Enzyme sweep daemon has been in continuous operation since approximately 2026-04-21 (the first entry in `logs/sweep-log.md`). The current operational record reflects ~3 weeks of production use across the wiki corpus.

Pass-level cost and latency

Table 1 summarizes per-layer and per-pass cost, latency, and token volume from representative sweeps during the operational window. Pass 2 dominates Layer 2 cost and latency because it reads the full corpus on every run; Pass 1 reads only the trigger and affected pages; Pass 3 reads the warm cache plus on-demand fetches. The corpus has grown over the operational window from 60 files (~560k Pass-2-input tokens, 2026-04-28, `logs/v4-synthesis-2026-04-28-3943bfc.md`) to 98 files (~866k Pass-2-input tokens, 2026-05-09, `logs/v4-synthesis-2026-05-09-259a8e8.md`). Pass 2 input includes the corpus plus prompt overhead, so Pass-2-input-tokens runs higher than the pure-corpus token count.

Table 1. Layer-and-pass cost, latency, and vendor assignment for the Open Enzyme two-layer cross-vendor architecture. All daemon costs in USD via OpenRouter. Layer 1 costs are subscription-based (Anthropic Claude Code) rather than per-call and are not enumerated per-edit. Representative values from sweep logs and `scripts/SWEEP-ARCHITECTURE.md` (`scripts/SWEEP-ARCHITECTURE.md`, `logs/v4-peer-review-2026-04-25-deepseek.md`).

Layer / Pass	Role	Model	Vendor	Typical cost	Typical latency	Input tokens (representative)
Layer 1 - Interactive	Origination: new content, cross-page connections, numbered computational-experiment briefs	Claude Opus 4.7 (via Claude Code)	Anthropic	Subscription; not per-call	Interactive, author-paced	Working sessions, variable
Layer 2 - Pass 1	Propagate	DeepSeek V4-Pro	DeepSeek	\$0.05–\$0.20	2–3 min	trigger + affected pages (~50–150k)
Layer 2 - Pass 2	Synthesize	DeepSeek V4-Pro primary; Google Gemini 2.5 Pro automatic fallback	DeepSeek / Google	\$0.40–\$0.80	5–7 min	corpus + prompt overhead: 559k–866k (2026-04-28 → 2026-05-09); Gemini run 2026-04-28: \$0.7322
Layer 2 - Pass 3	Review	OpenAI GPT-5.5 (default since 2026-05-08; Claude Opus 4.7 alternate)	OpenAI / Anthropic	\$0.05–\$0.30	1–2 min + tool calls	warm cache (triggers + cited files); on-demand fetches
Peer-review (episodic)	Independent cross-vendor verification	DeepSeek V4-Pro	DeepSeek	\$0.20–\$0.30	varies	full corpus; 2026-04-25 run: 467,964 in / 4,005 out, \$0.2070
Full Layer-2 three-pass sweep				\$0.50–\$1.30	9–12 min	

The daemon’s total monthly operational cost during normal use is in the low tens of dollars, a small fraction of the underlying API spend on a research project of this scope. The cross-vendor peer-review pass, when run episodically, adds approximately \$0.20–\$0.30 to that monthly total. Cost is not the binding constraint on the architecture; latency is, for use cases that need synthesis output in real time (see §7).

Verification-discipline costs (per-incident)

The daemon’s per-sweep costs (Table 1) sit alongside per-incident costs for the verification disciplines invoked when contamination or contention is suspected. Two such disciplines have quantified cost from the audit trail.

Brief-hygiene re-run. When a subagent brief is suspected of carrying contamination from a user-supplied framing example, the verification protocol is to re-run the same task with the contaminating phrase scrubbed and explicit anti-bias instructions added. The comp-018 → comp-020 verification (§5.2) cost approximately \$5 in subagent tokens plus 30–60 minutes of wall-clock time. The intervention surfaced three tied tier-1 candidates instead of the contaminated headline-of-one and reframed the central mechanism’s potency by 4–20×. Cost is cheap enough to make brief-hygiene re-runs default for any brief naming user-supplied compounds or concepts.

Verification-of-verification (V-of-V) chains. When a verification subagent’s output is suspected of regurgitating a laundered secondary-literature claim (§5.6), the protocol is to run an independent subagent tasked with checking the

first verification’s claims against primary source. The three-layer chain on the testosterone-axis adjuvant landscape (§5.6) cost approximately \$3–\$4 in subagent tokens. The intervention surfaced a non-existent supporting citation, a misattributed mechanism, and a direction-of-effect reversal in a wiki compound tag (GOUT-UNFAVORABLE → GOUT-FAVORABLE).

Both disciplines apply at the workflow level, independent of which models the passes use. The cross-vendor daemon catches one class of error (vendor-specific blind spots); these disciplines catch errors that survive cross-vendor review because the underlying contamination is upstream of the vendor.

Failure modes observed and recovery

Two distinct production failures during the 2026-04-28 window are documented in `scripts/SWEEP-ARCHITECTURE.md` (`scripts/SWEEP-ARCHITECTURE.md`) and motivated the operational hardening described in §3:

1. **Push-race failure (Run 25051936845, 2026-04-28).** Pass 1 propagated and pushed successfully. Pass 2 ran fully and produced a synthesis log locally on the runner. The runner’s push attempt was rejected because a concurrent commit (from Pass 1 of a parallel sweep) had landed on `main`. Pass 3 never ran. The synthesis artifact was lost when the runner tore down. Architectural response: rebase-before-push in every pass after Pass 1; failure ledger uploaded as a workflow artifact so traces survive runner teardown.
2. **API outage failure (Run 25049501442, 2026-04-28).** Pass 1’s OpenRouter call returned HTTP 503 (transient upstream outage). The shell wrapper exited non-zero on the first failed `curl`. No retry. The sweep terminated. Architectural response: retry-with-backoff (5s / 15s / 45s) wrapping every OpenRouter call; HTTP 5xx and curl exit codes 6/7/22/28 treated as retryable; HTTP 4xx treated as fatal (prompt or auth bugs, not transient outages).

A third failure class, Pass 3 admitting it could not verify a Pass 2 citation because the cited file was outside the trigger set, motivated the inter-pass artifact handoff described in §3 (the `propagated_files` and `cited_files` manifests).

Distribution of catches by class

Across the 13-day operational window 2026-04-25 to 2026-05-08, the five defensive case studies (§5.1–§5.4 + §5.6) are representative of five distinct failure classes; Figure 2 visualizes the catches alongside the surfacing mechanism for each. §5.5 is reported separately as a generative search-amplifier case rather than a defensive catch, and does not slot into the failure-class taxonomy. §5.1, §5.2, and §5.3 each surfaced exactly one instance during the window via within-pipeline discipline (manual walkthrough, independent re-run, ground-truth probe). §5.4 is the seminal cross-vendor catch that motivated the architecture, surfaced by DeepSeek V4-Pro’s independent peer-review pass on Claude-driven synthesis. §5.6 surfaced via an independent verification-of-verification chain targeting primary source, the only failure class in the taxonomy where cross-vendor heterogeneity does not provide a defense (the contamination distributes identically across vendors because it is inherited from training data). The operational data does not yet support a quantitative false-positive rate estimate; the production daemon has not flagged enough findings during the window to compute one with confidence. A more rigorous false-positive analysis is queued as future work once the operational record extends to ~6 months.

Figure 2. Catches by failure class during the operational window 2026-04-25 to 2026-05-08. Each cell represents one of the five failure classes; the bottom band’s color indicates the surfacing mechanism (indigo = cross-vendor surfacing via DeepSeek; grey = within-pipeline manual discipline; burgundy = verification-of-verification chain targeting primary source). The §5.4 catch is the cross-vendor catch that motivated the architecture and is the only one in the operational record surfaced by within-pipeline cross-vendor review. The §5.6 catch is the one failure class the cross-vendor pipeline cannot defend against (the contamination distributes identically across vendors), and is surfaced instead by independent V-of-V chains. Figure source: `figures/figure2_catches.py`.

The qualitative observation is that the most expensive catches by potential downstream impact are the cross-vendor methodological ones (§5.4 class). A single missed within-vendor cascade (§5.1 class) might propagate one wrong number across a dozen pages, recoverable. A missed methodological risk that aligns with the within-vendor model’s prior can shape the corpus’s direction for months before any reader notices.

Empirical inter-vendor agreement, pilot evidence

The case studies in §5 are qualitative and curated, six catches selected from approximately three weeks of operational deployment. A reviewer can reasonably ask whether the underlying premise of the cross-vendor heterogeneity

Figure 2. Failure-class taxonomy for the cross-vendor heterogeneity guard

Each cell is qualitative (N=1 in the 13-day operational window); the figure is a taxonomy of failure classes, not a frequency distribution.

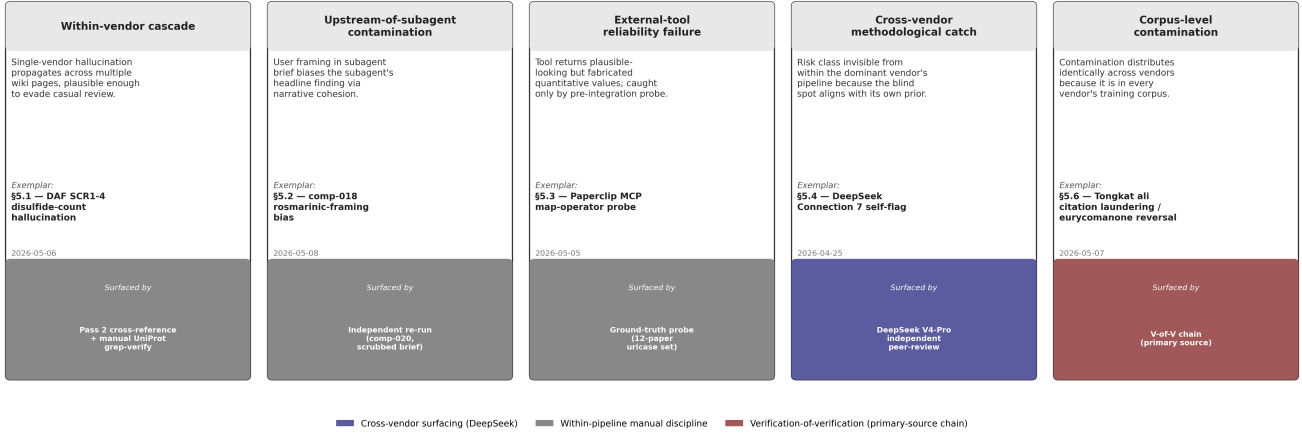


Figure 2: Figure 2. Catches by failure class

argument, that different-vendor frontier models actually produce meaningfully different outputs on the same scientific-synthesis prompts, is itself well-supported. This subsection presents pilot empirical evidence for that premise, drawn from a controlled study of eight prompts run across all four deployed vendors on 2026-05-14.

Method. Eight prompts spanning the daemon’s operational task types (one factual recall, one factual-quantitative, two mechanism-inference, one hypothesis-generation, one cross-document synthesis, one adversarial-review prompt taken near-directly from this manuscript’s own `review-prompts.md`, and one combined hypothesis-generation + mechanism task) were submitted to each of the four vendor models that participate in the deployed system: DeepSeek V4-Pro, Google Gemini 2.5 Pro, OpenAI GPT-5.5, and Anthropic Claude Opus 4.7. Each model was called via OpenRouter at the same temperature (0.7) with identical prompt text. Single-shot per (prompt, vendor) pair; no replicate sampling within-vendor. Claim coding was manual: for each prompt the human coder extracted five to seven atomic claims and assigned each vendor a verdict per claim (*affirm*, *disagree* with alternative recorded, *ignore*, or *refusal*). Pairwise agreement was computed across the six vendor pairs, restricted to claims where both vendors addressed the content substantively. The pilot ran for \$2.20; raw responses are preserved with SHA-256 prefixes at [papers/cross-vendor-heterogeneity-guard/empirical-vendor-comparison/](https://papers.cross-vendor-heterogeneity-guard/empirical-vendor-comparison/).

Results. Mean pairwise agreement across all 6 vendor pairs and all 8 prompts is **82.3%** (range across pairs: 76.7%-88.7%; range across task types: 70.0%-97.6%). All numerical results below are reported under the pilot’s manual claim-coding scheme, as small a study as N=8 prompts × 4 vendors permits.

Task type	n data points	Mean pairwise agreement	Min	Max
Factual recall	3	70.0%	50.0%	80.0%
Factual quantitative	6	97.6%	85.7%	100.0%
Mechanism inference	12	80.7%	66.7%	100.0%
Hypothesis generation	6	67.5%	50.0%	100.0%
Cross-document synthesis	6	87.5%	75.0%	100.0%
Adversarial review	6	78.6%	66.7%	100.0%
Hypothesis-gen + mechanism	6	89.7%	75.0%	100.0%

Vendor pair	n prompts	Mean agreement
DeepSeek vs Gemini	8	79.6%
DeepSeek vs OpenAI	8	88.7%
DeepSeek vs Anthropic	7	85.6%
Gemini vs OpenAI	8	79.6%
Gemini vs Anthropic	7	83.6%
OpenAI vs Anthropic	7	76.7%

The pattern is task-type dependent. Factual-quantitative recall is well-converged (~98% agreement) because the underlying literature consensus on numbers like ABCG2’s fraction of urate excretion, Q141K residual activity, and gut-vs-kidney attribution is mature and broadly cited across training distributions. Factual identity recall is much less converged (~70% agreement at N=3 data points), with most of the disagreement concentrated in the prompt-01 PTS1 finding documented below. Hypothesis-generation (~68% agreement) and adversarial review (~79% agreement) sit in between, where vendor priors most differ on which risks to foreground or which framing critiques to escalate.

Three findings worth flagging at the outset.

Finding 1: factual disagreement on a single three-letter answer (peroxisomal-targeting signal). Prompt 01 asked for the C-terminal PTS1 sequence of *Aspergillus flavus* uricase. DeepSeek answered SRL (Ser-Arg-Leu). Gemini and GPT-5.5 both answered SKL (Ser-Lys-Leu), the canonical answer per Legoux et al. 1992 (*J Biol Chem* 267:8565) and UniProt Q00511. Anthropic Claude Opus 4.7 refused the prompt entirely. Of three non-refusing vendors, two converged on the canonical SKL and one (DeepSeek) returned a near-neighbor wrong tripeptide. The point is not the 2-vs-1 split per se but what would have happened downstream of a single-vendor record. The comp-022 cassette-ranking analysis run 2026-05-14 as background project work recommends a PTS1-blocking C-terminal tag (3×Ala or His6) on the *A. flavus* uricase cassette; mechanistically, such a tag blocks PTS1 recognition by physically extending the C-terminus past whatever native tripeptide is there, so the *blocking strategy itself* is robust to the native-sequence identity. The failure surface is one step earlier: the wiki page documenting the cassette design would have inherited a wrong sequence annotation (recording “SRL” as the native PTS1 of *A. flavus* uricase), and any downstream reasoning that depends on the exact tripeptide (a residue-specific point-mutation alternative to physical tag blocking, a sequence-comparison argument against a homolog, a primer design that anneals across the C-terminus, a regulatory claim about Pex5-binding affinity) would have inherited the error. This is the failure mode the heterogeneity guard is designed to catch at the corpus-content layer, demonstrated on the pilot’s first prompt. The PTS1 finding also turned out to have a reflexive audit-trail dimension, returned to in the meta-catch paragraph below.

Finding 2: a vendor caught a planted factual error in the prompt itself. Prompt 05 contained an embedded factual error describing pegloticase as “mammalian-cell-produced.” Actual pegloticase is produced in genetically modified *E. coli* and then PEGylated. OpenAI GPT-5.5 explicitly pushed back with a “Minor factual correction” before answering. DeepSeek accepted the prompt’s framing without flagging. Gemini accepted the framing and built it out into a mechanism claim about glycan structure (“produced in mammalian cells, resulting in more human-like complex glycan structures”) that compounds the error. Anthropic accepted the framing with ambiguous hedging but did not push back. This is a different mechanism than Finding 1, not divergent recall of an unstated fact, but divergent willingness to push back on a stated framing. The cross-vendor heterogeneity guard captures both.

Finding 3: Anthropic uniquely surfaced a critical engineering mechanism (cns3) that the other three vendors missed. Prompt 08 (cordycepin co-expression feasibility) asked vendors to evaluate adding cns1+cns2 to a uricase+lactoferrin koji cassette. The known *Cordyceps militaris* biosynthetic gene cluster includes a third gene, cns3, a pentostatin-like adenosine-deaminase inhibitor that confers self-protection on the producer host. All four vendors flagged ADA-mediated cordycepin conversion as a risk. Only Anthropic named cns3 as the solution and called it “non-negotiable”; DeepSeek, Gemini, and GPT-5.5 each proposed ADA knockouts or downstream engineering instead. This connects directly to comp-025 (ADA pool competition, queued 2026-05-14 in the project’s [wiki/computational-experiments.md](#) §Planned Analyses): Anthropic had already named the canonical resolution the comp brief is being written to scope. A single-vendor pipeline operating on DeepSeek, Gemini, or GPT-5.5 alone would have proposed a substantially weaker cassette design.

The PTS1 disagreement (factual) and the cns3 catch (mechanism-completeness) instantiate two of the failure modes §4 names: the cumulative homogenization risk for shared-prior facts, and the search-amplifier value where one vendor surfaces a mechanism the others do not propose unprompted.

Anthropic refusal as a separate failure-mode signal. Anthropic Claude Opus 4.7 refused prompt 01 entirely (`native_finish_reason: refusal`) and produced full content on the other seven prompts. The refusal reproduced across multiple wordings of the same question. This is a vendor-specific brittleness on certain prompt phrasings, an interesting heterogeneity-guard finding in its own right: a single-vendor pipeline routed through Anthropic alone would have had a coverage gap on this class of prompt that no within-vendor review would have surfaced. From the cross-vendor pipeline’s perspective, the other three vendors covered the gap and the loss was absorbed at the architecture level rather than the output level.

Partitioning the disagreement: within-vendor replicate study. A natural next question is how much of the

17.7% cross-vendor disagreement is real vendor-prior difference versus within-vendor temperature stochasticity that would produce similar disagreement on consecutive calls to the same model. To partition the variance, each of the 32 (prompt, vendor) cells from the original pilot was called two additional times at the same prompt, same vendor, same temperature (0.7), same max_tokens cap. Each replicate response was coded against the same atomic-claim taxonomy from the original run. Within-vendor pairwise agreement is computed across the three replicates per cell (r1-r2, r1-r3, r2-r3 = three pairs), aggregated across the 32 cells. The replicate study cost \$4.38 incremental (total pilot + replicates: \$6.58).

Headline within-vendor agreement rate: approximately 90.8%, compared to the approximately 82.3% cross-vendor headline. The within-vendor rate is about 8.5 percentage points higher than the cross-vendor rate. Interpreting the gap as a partition under the pilot’s manual coding scheme: cross-vendor disagreement ~17.7%; within-vendor disagreement (temperature noise floor) ~9.2%; residual cross-vendor heterogeneity signal ~8.5% (roughly half of total cross-vendor disagreement). About half of the original cross-vendor disagreement is real prior-distribution difference between vendors; the other half is within-vendor noise that any single-vendor pipeline running at the same temperature would also produce on a second call. **The architectural-principle claim survives the partition but at roughly half the magnitude that the headline cross-vendor disagreement would suggest if interpreted naively.** The numerical decomposition is intended as a pilot-scale signal, not a clean biological-style variance decomposition: N=8 prompts with hand-coded atomic claims, single coder, three replicates per cell. The qualitative finding (the cross-vendor signal is real and roughly comparable in magnitude to within-vendor noise at this temperature) is robust to the exact numbers shifting with a larger replicate or with a second coder; the precise partition fraction is not.

The per-vendor structure of within-vendor stability does *not* support the reasoning-mode hypothesis suggested by the original pilot. The earlier interpretation speculated that DeepSeek V4-Pro and OpenAI GPT-5.5 might be individually more stable across replicates than the non-reasoning Gemini and Anthropic models, which would have explained why DeepSeek-vs-OpenAI was the highest-agreement cross-vendor pair (88.7%). The replicate data falsify that explanation:

Vendor	Within-vendor agreement (replicate study)	Cross-vendor agreement (pilot)
Anthropic Claude Opus 4.7	94.0%	81.9%
DeepSeek V4-Pro	90.7%	84.6%
Gemini 2.5 Pro	89.9%	80.8%
OpenAI GPT-5.5	88.3%	81.9%

Anthropic is the *most* stable vendor across replicates, not the least; the non-reasoning models (Anthropic 94.0%, Gemini 89.9%) bracket the reasoning models (DeepSeek 90.7%, OpenAI 88.3%) on both sides. The DeepSeek-vs-OpenAI cross-vendor agreement (88.7% in r1) is better explained as a real prior overlap between the two reasoning-mode pipelines than as an artifact of within-vendor stability. Anthropic’s high within-vendor rate is partly inflated by its refusal stability on prompt 01 (the model refused all three replicates, and “both refused” is coded as substantive agreement); excluding that effect, the four vendors are within ~7 percentage points of each other in within-vendor stability.

The replicate study sharpens two of the three pilot findings:

The PTS1 prompt shows partial within-vendor stability and partial within-vendor drift. Across the three replicates of prompt 01, DeepSeek answered SRL (r1) then SKL (r2) (changed across replicates); Gemini answered SKL twice (r1, r2) then AKL (r3) (stable at the canonical answer for two replicates, then drifted at r3); OpenAI answered SKL (r1) then AKL (r2) then SKL (r3) (drifted once and recovered). Anthropic refused all three replicates (the most stable single behavior on this prompt). Two of three non-refusing vendors gave at least one wrong answer across the replicate set, and two of three changed their answer at least once. A single-vendor pipeline running at the same temperature would produce different PTS1 answers on subsequent calls, sometimes the canonical SKL and sometimes a near-neighbor wrong answer (SRL, AKL). The cross-vendor guard’s value here is not only that vendors disagree with each other but that they disagree with themselves across replicates; the guard surfaces the underlying uncertainty regardless of which vendor produced the seed answer.

The cns3 catch is stable across replicates. Anthropic uniquely named cns3 as a non-negotiable resistance gene in all three replicates of prompt 08. DeepSeek, Gemini, and GPT-5.5 again did not name cns3 as a kill-criterion-level requirement in any of r1, r2, or r3. The cross-vendor catch is reproducible: a stable vendor-prior difference, not a within-vendor coincidence. This is one of the cases the partition cleanly preserves as real cross-vendor heterogeneity.

The pegloticase catch shows a mixed pattern. The catching-side stability is asymmetric (OpenAI catches the planted error all three replicates; Anthropic catches r2+r3; DeepSeek catches only r3; Gemini misses all three and extends the error into the same human-like-glycan mechanistic claim each time). Gemini’s replicate-stable pattern of accepting AND amplifying the error across three calls is more concerning than a one-off error because it indicates the failure mode is structural to the vendor’s response prior rather than a random sample.

A further reflexive audit event during manuscript preparation. While this manuscript was in final review, an independent cross-vendor audit pass (a different model family from the one that drafted §6) compared the prose of the PTS1 finding against the underlying response artifacts in `empirical-vendor-comparison/responses/01-uricase-mechanism-factual/`. The audit surfaced a provenance mismatch: the manuscript prose and the C1.5 claim-coding file both recorded Gemini’s r1 answer as “SSH (Ser-Ser-His)” disagreement, while the raw stored response `gemini-r1.md` contains the canonical “SKL (Ser-Lys-Leu)” affirmation. The mismatch had passed through the empirical-vendor-comparison subagent’s claim coding, through orchestrator integration into §6, through two prior hostile-persona review passes that focused on the prose and the architectural framing rather than on artifact-level reconciliation. It was detected only when an independent model family was asked to verify the manuscript’s empirical claims against the raw response files. We corrected the claim-coding file (preserving the original wrong coding as an `audit_correction` field so the trail remains inspectable), re-ran the agreement-matrix analysis, updated the pilot-headline numbers reported above (cross-vendor 82.3% rather than the previously reported 81.8%; factual-recall task type 70.0% rather than 63.3%; per-pair Gemini-vs-OpenAI 79.6% rather than 77.1%), and rewrote the Finding-1 narrative in this section to reflect the corrected 2-of-3-correct rather than 3-of-3-different outcome. The architectural-principle claim and the partition magnitudes are robust to the correction (the within-vendor replicate study’s headline shifted from 90.6% to 90.8%; the real-heterogeneity gap from 8.7 pp to 8.5 pp). The qualitative finding is unchanged: vendors disagree on PTS1, and a single-vendor pipeline using DeepSeek’s SRL would have engineered against the wrong sequence. The event does not prove that the audit vendor is “better” than the drafting vendors, and it does not prove cross-vendor review catches everything. It does show that vendor-independent review can catch errors that survive a strong same-vendor author-review-and-review pipeline, and it argues for keeping raw artifacts, claim codings, and manuscript claims inspectable as separable layers rather than collapsing them into a single edited surface. This is the same central claim the rest of the paper makes, instantiated again at the manuscript-production scope, with the scar visible in the audit trail rather than hidden by the correction.

Interpretation. The pilot data plus the replicate-study partition are consistent with the architectural-principle claim that motivates the heterogeneity guard, at roughly half the magnitude the headline cross-vendor disagreement would suggest if interpreted naively. The observed ~17.7% cross-vendor disagreement decomposes into ~9.2% within-vendor temperature noise and ~8.5% real cross-vendor heterogeneity under the pilot’s manual claim-coding scheme; the cross-vendor signal is real and reproducible but is not the entirety of the original headline. The disagreement is concentrated in the task types the daemon’s value most depends on: factual identity recall (where the corpus depends on the vendor getting the right molecular identity), hypothesis generation (where the vendor’s choice of which risk to foreground shapes downstream experimental design), and adversarial review (where catch-count differences directly determine which weak points get flagged). The PTS1 finding shows that single-vendor pipelines are unstable even against themselves on the same prompt at the same temperature; the cns3 finding shows that a real cross-vendor prior difference is reproducible across replicates. The §5 case-study evidence and the §4 architectural argument carry the rest of the inference.

The pilot’s findings sit inside a published literature on inter-vendor agreement that broadly supports the mechanism claim and bounds it with one important caveat (frontier convergence at scale). The next subsection situates this pilot against that literature.

Limitations. Five load-bearing items, each disclosed in the pilot’s reproducibility artifact (`empirical-vendor-comparison/`):

1. *N=8 prompts is small.* Variance on per-task-type mean agreement is wide; a controlled study at N=50-100 prompts is appropriate future work.
2. *Within-vendor variance was directly measured.* The pilot originally called each cell once; a follow-up replicate study added two more calls per cell at the same temperature, producing three replicates per cell across all 32 cells (64 additional API calls). The partition (within-vendor ~90.8%, cross-vendor ~82.3%) splits the headline cross-vendor disagreement roughly evenly between real cross-vendor heterogeneity and within-vendor noise; the architectural-principle claim survives but at roughly half the magnitude the headline cross-vendor disagreement

would suggest if interpreted naively. $N=3$ replicates per cell captures modal behavior; $N=10+$ per cell would tighten the per-vendor stability estimates and is appropriate future work.

3. *Manual claim extraction, single coder.* Claim verdicts are auditable per-claim in `outputs/claims/*.json`, but no second-coder cross-validation pass was performed.
4. *Synthetic-but-realistic prompts.* Seven of eight prompts mirror the daemon’s operational task shape rather than being drawn directly from production logs. One (prompt 07) is taken near-directly from this manuscript’s own `review-prompts.md`.
5. *The pilot does not measure single-vendor sufficiency.* It measures whether vendors disagree on the same prompts (a necessary condition for the guard to add value), not whether a single-vendor pipeline-plus-rigorous-self-review would catch the same disagreements. The controlled ablation comparing cross-vendor against single-vendor-with- N -self-review-passes is queued as future work in §7.

Inter-vendor agreement in the published literature

The pilot evidence above is the first measurement we are aware of for cross-vendor disagreement on iterative wiki synthesis with cross-document propagation. The published literature on inter-vendor LLM agreement covers adjacent task types (multiple-choice reasoning, clinical diagnosis, general-benchmark accuracy) and broadly supports the mechanism the heterogeneity guard relies on, with one important caveat.

Empirical inter-vendor disagreement is measurable and substantial. Amiri-Margavi et al. 2025 (arXiv:2411.16797) report Fleiss’ kappa on 100 PhD-level statistics questions ranging from 0.7160 (Claude, substantial agreement) to 0.2811 (Gemini, fair agreement) across GPT-4, Claude-3-Opus, Gemini-1.5-Flash, and LLaMA-3-70B. Full-agreement rates spanned 86% (Claude-generated) to 65% (LLaMA-generated). A different domain (graduate statistics versus our biology), a different metric (consensus rate versus our pairwise agreement), and the same conclusion: distinct vendors produce distinctly different outputs on scientific reasoning, and the disagreement is large enough to be statistically meaningful.

The disagreement is exploitable for accuracy lift. Yuan et al. 2025 (arXiv:2603.04421) report a 10-point absolute recall@1 lift on the hardest rare-disease diagnostic subset (40 cases) when a mixed-vendor team of o1-mini + Gemini-2.5-Pro + Claude-4.5-Sonnet replaces the best single-vendor model. The lift is attributed to “complementary inductive biases that homogeneous teams could not pool” and explicitly inversely correlates with Jaccard similarity of vendor outputs (range 0.269–0.750 on the test set). This is the precise mechanism the present paper proposes: low cross-vendor overlap predicts large rescue gains.

The architectural pattern is independently validated. Verga et al. 2024 (arXiv:2404.18796) introduce a Panel of LLM Evaluators using juries of “disjoint model families” and find that they reduce intra-model bias relative to single-judge baselines while running at $7\times$ lower cost. Zhang et al. 2025 in a meta-analysis of multi-agent debate methods conclude that homogeneous-model debate often fails to beat single-agent baselines, and that “model heterogeneity acts as a universal antidote” to homogenized-model failure. Their position-paper recommendation is that the field “actively embrace model heterogeneity as a core design principle.” The present paper’s daemon architecture instantiates this principle for a specific class of scientific-synthesis task.

The caveat: frontier convergence. Kim et al. 2025 (ICML, arXiv:2506.07962) survey 350+ LLMs and report that frontier LLMs agree 60% of the time when both make errors on a shared leaderboard task. Two implications follow. First, the 60% co-error rate is well above chance, so vendors share substantial error structure, particularly within a provider family or architecture. Second, the 60% is well below the 100% co-error rate a fully homogenized ecosystem would produce. The 40-point gap is the headroom the heterogeneity guard can exploit. The pilot data above sit inside that gap. The more important Kim 2025 caveat is that **larger and more accurate models have highly correlated errors even with distinct architectures and providers**. Model size has a negative impact on epistemic diversity. As frontier models converge on the same training corpora, the same RLHF objectives, and the same benchmark-optimized reasoning patterns, their errors converge too. This does not invalidate the mechanism claim; it bounds it. The heterogeneity guard catches errors that exist in the cross-vendor variance budget. As that budget shrinks at the frontier, the guard’s marginal value shrinks with it. The honest framing of the contribution is therefore not “vendor diversity solves the homogenization problem” but “vendor diversity is one of the few mechanisms still available to recover error coverage in a frontier-converging ecosystem.”

What the literature does not yet answer. Three gaps the present pilot addresses or begins to address: no published study, to our knowledge, measures cross-vendor disagreement specifically on iterative wiki synthesis or multi-pass biological reasoning over a connected knowledge graph; no study disentangles agreement-on-task-accuracy from agreement-on-outputs (two vendors that get the right answer for different reasons register as agreeing on most

existing metrics, but their outputs diverge on the reasoning, which is where the present pipeline mines value at Pass 3 critique and the episodic peer-review pass); and no controlled study tests DeepSeek-family models in the same cross-vendor agreement framework as the major Western-vendor offerings. The pilot above contributes to the first two; the third is partially addressed by including DeepSeek V4-Pro in the four-vendor panel.

§7 Limitations and failure modes

The cross-vendor heterogeneity guard described in this paper is not a complete defense against AI-assisted-synthesis failure modes. This section enumerates the failure classes the architecture does *not* protect against, plus the operational costs.

What the architecture does not protect against

Shared training-data leakage. All major frontier model vendors train on substantially overlapping web crawls. Common-cause biases from the underlying training corpora (e.g., over-representation of English-language scientific literature, under-representation of certain non-Western research traditions, blind spots in pre-2020 scientific consensus that has since shifted, and laundered claims from secondary-literature tiers that every major model treats as common knowledge) are not eliminated by switching vendors. The cross-vendor pattern guards against *vendor-specific RLHF and alignment biases*, which are real and significant, but it cannot guard against shared upstream-data biases. §5.6 is the worked example of this class: a multi-layer citation-laundering chain (the “37% testosterone elevation” figure for *Eurycoma longifolia*, a non-existent supporting citation, and a wrong mechanism attribution that flipped a wiki tag from GOUT-UNFAVORABLE to GOUT-FAVORABLE) defeated cross-vendor review because every vendor’s training corpus carried the same contamination. The defense that worked was independent verification chains targeting primary source, distinct in mechanism from cross-vendor heterogeneity. Mitigations against this class are documented separately in `CLAUDE.md` §“Global-multilingual research by default” and §“Translation protocol.”

Adversarial prompt injection. Nothing in the cross-vendor architecture defends against an adversary who can inject instructions into the wiki content itself. The daemon will faithfully propagate, synthesize, and review whatever it reads. Defense against this class requires content-source authentication, which is outside the scope of this paper.

Catastrophic model failure modes. If one vendor’s model emits content that is obviously wrong by any reasonable read (e.g., a factual contradiction at the level of the local sentence), the cross-vendor pass will catch it. If one vendor’s model emits content that is *subtly* wrong in a way the adjacent vendor’s model shares the same blind spot on, the pattern provides no defense. The base-rate frequency of such shared-blind-spot failures is unknown and likely depends on which specific vendors are deployed.

Operator error. The five operational hardening components described in §3 catch most operator-error classes (typos in commit markers; missing artifact handoff; race conditions). But operator decisions about which models to deploy on which passes still matter; collapsing the pipeline to a single vendor for cost reasons would defeat the architecture.

Prompt brittleness across model releases. Each pass’s logic is encoded in a prompt, and prompt behavior is vendor-tied. The driver of silent pipeline degradation in the operational record is not silent vendor updates to incumbent models but new model releases that require vendor-specific prompt tuning. When GPT-5.5 launched, the initial Pass 3 eval ran the incumbent Anthropic-tuned prompt against the new model and the result looked worse than Opus 4.7 on the same synthesis log. A too-narrow eval would have stopped there. Reading OpenAI’s published prompt-engineering guidance for GPT-5.5 and re-tuning the prompt to match (outcome-first ordering, decision rules in place of ALWAYS/NEVER, explicit retrieval triggers) flipped the result: GPT-5.5 then produced the best Pass 3 verdicts the project had seen, at $\sim 5\times$ lower cost and with deeper per-marker verification (2 $\rightarrow$ 7 tool-using rounds; evals/pass-3-reviewer/2026-05-07-abc8de9-comparison.md, commit e728d0d).

The lesson: cross-vendor heterogeneity does not survive “swap the model and re-run.” It requires per-vendor prompt tuning aligned to each vendor’s published guidance, plus evals broad enough to distinguish “this model is worse” from “our prompt is worse for this model.” Without the GPT-5.5 retune, the project would have correctly concluded the model was inferior on the existing prompt and incorrectly concluded that cross-vendor heterogeneity was hard to scale. The brittleness is real; the defense that works is empirical eval discipline plus vendor-guidance reading, not infrastructure.

Operational costs

Cost overhead vs. single-pass. A three-pass cross-vendor sweep runs approximately $3\text{--}4\times$ the API spend of a single-pass single-vendor synthesis on the same trigger. The Open Enzyme deployment averages a few dollars per sweep at current OpenRouter pricing; for higher-volume deployments, the multiplicative cost is the primary tradeoff.

Latency overhead. End-to-end sweep latency is 9-12 minutes per the operational data in §6, vs. seconds for a single-pass synthesis. For workflows where downstream readers need synthesis output in real time, the cross-vendor pattern is too slow. The pattern is designed for *background* synthesis on a slowly-evolving corpus, not for interactive query-response.

Operational complexity. Five hardening components, three sets of pass prompts, a state registry, hooks, and a watchdog cron are non-trivial infrastructure for a small research project. The pattern is justifiable when the corpus is consequential for downstream decisions (research direction, wet-lab spend, hypothesis commitment); it would be over-engineered for ephemeral synthesis tasks.

When this matters and when it doesn't

The cross-vendor heterogeneity guard is most valuable when:

- The corpus is **long-lived** (not a one-shot query) and accumulates over many edits.
- The corpus is **consequential** for downstream decisions (research direction, regulatory submissions, clinical claims, financial allocation).
- The cost of a single class of systematic error (within-vendor blind spot) is high relative to the operational cost of the pattern.
- The synthesis problem is **literature-heavy** (vs. data-heavy), so the relevant failure mode is misreading the literature, not miscomputing on data.

The pattern provides little value when:

- The query is one-shot and the output is consumed by a human who will verify it directly.
- The cost of the additional passes exceeds the cost of the errors they would catch.

The pattern's value is reduced, though not eliminated, when the task is purely numerical and solvable with deterministic code. For hybrid tasks involving both literature synthesis and extraction of reported quantitative claims (clinical trial summaries, kinetic-parameter compilations, dose-response surveys), the cross-vendor pass retains meaningful value: it can catch hallucinated numbers, misquoted sample sizes, and mis-attributed kinetic parameters that within-pipeline review systematically misses. The §5.3 Paperclip probe in this paper is an instance of exactly this, a hybrid literature/numerical task where the cross-vendor pass surfaced kinetic-parameter misreports that the source-tool's within-vendor pipeline had not flagged.

§8 Discussion

How many vendors is enough?

The pattern described in this paper uses three vendor pipelines (Anthropic, DeepSeek, OpenAI; with Google Gemini as a configured alternate at Pass 2) across three operational passes, plus episodic peer-review from a separate vendor. Whether three is the right number is an empirical question. Two vendors would provide a heterogeneity guard against the most common form of within-vendor blind spot but would lose the diversity that comes from having a third perspective break ties. Four or more vendors would provide additional independence at additional cost and operational complexity. The Open Enzyme deployment landed on three as a working compromise; replication studies at different scales would refine this.

A subtler question is whether the choice of *which* vendors matters as much as the *number*. Two vendors with substantially overlapping training data (e.g., two frontier models that both train heavily on Common Crawl with similar RLHF objectives) may be effectively within-vendor for the purposes of heterogeneity. Two vendors with deliberately divergent training pipelines (e.g., a Western frontier model and a Chinese frontier model) likely provide more heterogeneity per dollar. The current Open Enzyme assignment includes a Chinese-vendor model (DeepSeek) precisely for this reason, though the empirical evidence for the heterogeneity gain is anecdotal at this scale.

Does the guard scale to non-text modalities?

The pattern as described operates on text corpora. Whether it generalizes to vision (e.g., medical-image synthesis), structure prediction (e.g., AlphaFold-class protein-structure synthesis), or multi-omics (e.g., genomics-plus-proteomics integration) is an open question. The conceptual argument, that single-vendor blind spots accumulate over many edits, should apply across modalities, but the operational pattern (read full corpus, emit synthesis, review with verdicts) is text-specific. A vision or structure analog would require its own operational pattern; the heterogeneity-guard principle should survive the translation.

The cross-vendor pattern and the human in the loop

A natural reviewer question at this point: is the cross-vendor architecture doing the work, or is the human reviewing its output doing the work? Both, and the architecture’s value depends on that division. On the safety-net side, the cross-vendor passes narrow the surface a human must read from “the entire corpus” to “the verdict-tagged synthesis-queue items.” On the search-amplifier side, the passes widen the perspective by surfacing candidate findings the dominant-vendor pipeline alone would not have proposed. A human triages both kinds of output via the **synthesis/queue/**walkthrough, applying judgment about what to action, what to defer, and what to drop. None of the §5 catches were fully autonomous (§5.1 was sweep + manual UniProt walkthrough; §5.2 was a human-specified rerun; §5.3 was a human-curated ground-truth probe; §5.4 was a machine-surfaced observation that a human read, recognized as consequential, and elevated to architectural commitment; §5.5’s three convergent findings were generated by the AI passes and actioned by a human across the subsequent two weeks). The §5.5 closing observation about the 2026-05-08 incident points to a second structural property: in some convergent generative moments the human is one of the vendors in the convergent pair, a Gemini-driven synthesizer and a human writer can independently surface the same external-comms angle, which is the same heterogeneity pattern at a different substrate level. This manuscript’s own audit trail (Appendix B) likewise records several human-only catches the cross-vendor review pass missed. The cross-vendor pattern is therefore best understood as both a *trigger-and-filter* and a *search-amplifier* for human attention, not as a replacement for human judgment, which places it closer to PaperOrchestra (human-in-loop for the science, automation for synthesis-and-drafting) than to AI Scientist v2 (full-loop automation) on the spectrum surveyed in §2.5.

Implications for autonomous-research systems

A class of recent systems aspires to **end-to-end automated scientific research**: hypothesis generation → experiment design → analysis → manuscript drafting, with humans optionally involved at any step (<https://0xbpa.substack.com/p/grounding-the-ai-scientist-hype>). For systems where each agentic loop iteration depends on the previous step’s AI output, the epistemic homogenization risk compounds: any blind spot in the system’s prior can propagate across hundreds of synthesized hypotheses before any human notices. The cross-vendor heterogeneity guard described here is one tractable defense against that compounding. Autonomous-research systems that deploy a single vendor across all stages are accumulating the same vendor’s prior at every loop iteration; the corpus drift is the system’s drift.

The reflexive observation: this manuscript was drafted using the methodology it describes. Cross-vendor review fired at section boundaries during drafting (DeepSeek on §4 and §5; Gemini on §3, §6, §7; Claude Opus 4.5 plus DeepSeek in parallel on §2). The production process is documented in the Methods Appendix; Appendix B’s working log records on the order of 50 logged catches across the full drafting and review history (initial cross-vendor passes, the original Codex review pass, the wide-scope audit, two hostile-persona review rounds, the within-vendor replicate study, the final cross-vendor audit-trail reconciliation that produced the §6 meta-catch documented above, and final figure / scope / number-cleanup catches). Roughly 40% are author-or-self-caught and 60% are surfaced by independent reviewers; the exact attribution is in **revisions.md**. The catches map onto the failure classes §5 enumerates: a primary-drafter confabulation caught by user push-back (§5.1 within-vendor cascade class, surfaced during the drafting of the very case study that warns against it); a §7 limitation paragraph that fabricated a “silent vendor update” risk the operational record does not show, replaced via author read-pass with the actual GPT-5.5 launch + retune story (analogous to §5.6: the contamination is a plausible-sounding generalization rather than a vendor-specific bias); a stale Pass 3 model name that survived the original drafting until a self-verify pass; a Figure 1 that read as spaghetti and required two redraw passes. The audit trail’s value is showing what the methodology caught and what it missed when applied to the paper’s own production.

§9 Conclusion

Cross-vendor heterogeneity is a tractable engineering pattern for AI-assisted scientific synthesis. By routing successive passes of a synthesis pipeline through models from different vendors, not just different models from the same vendor, a research project can guard against the accumulation of any single vendor’s blind spots and biases in a long-lived corpus.

The Open Enzyme deployment is one instance of the pattern, operating on a single research wiki for an open-source biotechnology project. The architecture (three operational passes plus five hardening components plus an episodic peer-review pattern), the seven case studies (a reflexive §5.0 leading example, four defensive operational catches across §5.1–§5.4, one search-amplifier generative case in §5.5, and a boundary-case limitation in §5.6 that the complementary verification-of-verification discipline catches rather than the architecture itself), and the pilot empirical study with within-vendor partition (§6) are concrete enough to replicate. The code is open. The logs are open. The audit trail is open. The pattern generalizes.

The structural argument is simple: in long-lived AI-assisted synthesis, the diversity that protects the corpus has to live somewhere outside the corpus. Cross-vendor review is one place to put it. Within-vendor review, by construction, is not.

Methods Appendix, Cross-vendor production process used for this manuscript

This appendix documents the production process for the manuscript itself, in keeping with the reflexive note in §8.

Primary drafter, §3–§9: Anthropic Claude Opus 4.7, working from the Open Enzyme wiki primary sources, the sweep daemon code and prompts, the operational logs, and the engineering documentation. Drafting and revision spanned multiple working sessions on 2026-05-13, captured chronologically in the audit trail (Appendix B). The final manuscript reflects: initial section drafting, cross-vendor review at section boundaries (DeepSeek + Gemini + Claude reviewer in parallel), response-to-review revision, a primary-source grep-verify pass on load-bearing claims, an external Codex review pass and response, a wide-scope audit of the project’s catch history for omitted material (`audit-2026-05-13-catch-history.md`), and a final author read-pass that surfaced additional catches not visible to any of the prior reviewers.

Primary drafter, §2 related work: Anthropic Claude Opus 4.7 driving the **Ar9av/PaperOrchestra community implementation** of the PaperOrchestra writing pipeline (the Song et al. 2026 paper itself ships no code; Ar9av at github.com/Ar9av/PaperOrchestra is the only known usable implementation, released as a skill pack for coding agents). Inputs: this paper’s outline, the existing §3–§9 drafts, and wiki primary sources as the “rough notes and experimental logs” the pipeline ingests. The pipeline’s Literature Review agent verifies cited papers exist via Semantic Scholar; the host LLM (Claude) does all reasoning work, which means §2 is Anthropic-driven through and through. The §2 cross-vendor review pass (Claude Opus 4.5 and DeepSeek V4-Pro in parallel) is the architectural step that compensates for this.

Cross-vendor review at section boundaries: §4 and §5 were reviewed by DeepSeek V4-Pro independently after initial drafting; §3, §6, and §7 by Google Gemini 2.5 Pro; §2 by Anthropic Claude Opus 4.5 and DeepSeek V4-Pro in parallel. All review passes were orchestrated via OpenRouter and the verbatim model outputs are preserved as supplementary material. Catches from each review pass are logged in the audit trail (Appendix B).

Primary-source grep-verification pass: the drafter applied the project’s own pre-commit grep-verify discipline (the rule articulated in §5.1) to load-bearing quantitative and identity claims in §4 and §5 before commit. Catches from this self-verification pass are recorded in the audit trail (Appendix B).

A reflexive catch worth surfacing here: early in drafting, the author asked whether Google’s PaperOrchestra should be used to draft the paper. The primary drafter responded with a confabulated “I don’t have reliable information on Paper Orchestra” without checking the project’s own repository, which contained two files (`Open Enzyme/posts/notes/hypothesis-generation-gap.md` and an essay-form analysis of the AI-scientist landscape) that described PaperOrchestra in detail. The catch came from the author pushing back (“did you look in the repo?”). This is the same failure class as §5.1 (within-vendor confabulation that should have been caught by primary-source verification), surfaced during the drafting of the very case study that warns against it. It is the first entry in the audit trail (Appendix B) and is preserved here as the cleanest available demonstration that the heterogeneity guard has to apply to the paper’s own production process.

Appendix A, Vendor attribution by section

Reading note. The “Independent reviewer” column records the *initial* cross-vendor review pass per section (DeepSeek V4-Pro or Google Gemini 2.5 Pro via OpenRouter, fired at section boundaries during original drafting). In addition, **the entire manuscript subsequently received four independent Codex review passes** (OpenAI-family commercial coding/research agent, hostile-persona prompted): an initial Codex audit, two hostile-persona review rounds (second-pass + third-pass), and a final-check pass on 2026-05-14 that surfaced the PTS1 / Gemini-r1 audit-trail mismatch documented in the §6 meta-catch box and regenerated both figures. Because Codex review applied at the whole-manuscript level rather than at section boundaries, every section saw at least one Codex review pass even where the initial cross-vendor pass was an author read or skipped. The per-row entries below now credit Codex where its review specifically surfaced a catch or drove a section-level change; the broader manuscript-level Codex coverage is implicit across all rows.

Section	Primary drafter	Independent reviewer	Catches logged
Abstract	Anthropic Claude Opus 4.7	Codex (four review passes; second-pass surfaced “abstract overlong” → ~510 word → 360 word trim per reviewer’s outline; final-check pass surfaced the post-PTS1-audit number updates)	Catch 42 (abstract trimmed: problem / method / deployment / strongest catch / pilot / limitation outline per second-pass Codex reviewer); Catch 49 (“literature” dropped from title per author + reviewer agreement); Catch 50 (partition headline numbers updated after PTS1 audit-trail catch)
§1 Introduction	Anthropic Claude Opus 4.7	Codex (across four review passes; drove “vibe-science” residue scrub, “five case studies” → “seven case studies + leading reflexive example” reframing, “strictly stronger” softening, “literature synthesis” → “synthesis” scope tightening)	Catch 41 (mechanical sweep: vibe-science residue + five-case-studies residue); Catch 44 (“strictly stronger” → “adds a capability not supplied by verification alone”); Catch 49 (“literature” dropped from title and paper-scope framings)
§2 Related work	Ar9av/PaperOrchestra community implementation, driven by Claude Opus 4.7 (Path B install)	Anthropic Claude Opus 4.5 + DeepSeek V4-Pro (parallel; via OpenRouter) on the initial pass; Codex second-pass and third-pass reviews surfaced four new citations (Kim 2025, Yuan 2025, Amiri-Margavi 2025, Zhang 2025) and corrected paraphrased reference titles	Catch 6 (S2 rate-limit, RESOLVED same-day after key arrived); Catch 7 (PoLL vendor-composition mischaracterization, self-caught); Catch 8 (AI Scientist v2 venue-rigor inflation, self-caught); Catch 10 (§2.4 Anthropic-skewed citation selection, InstructGPT added); Catch 17 (FARS reference dropped, unsupported by verified citation); Catch 18 (Yamada 2025 “first workshop” softened, Lu 2024 precedent); Catch 19 (Du et al. mixed-model framing); Catch 20 (Shumailov preprint+published clarified); Catch 43 (four new citations added per second-pass Codex literature-grounding ask); Catch 45 (reference accuracy: refs 17–20 paraphrased titles corrected against canonical arXiv strings per Codex third-pass nit). 2 false positives from Claude reviewer (date confusion).

Section	Primary drafter	Independent reviewer	Catches logged
§3 Architecture	Anthropic Claude Opus 4.7	Google Gemini 2.5 Pro (via OpenRouter) + author read-pass + audit-driven additions; Codex final-check pass regenerated Figure 1 (vibe-science → “Interactive hypothesis development”, comp-NNN shorthand replaced)	Catch 12 (peer-review-pass vendor collision with Pass 2, assignment reframed); Catch 13 (Gemini “as fallback” under-credit, reframed as full primary backend); Catch 32 (DeepSeek/Gemini Pass 2 routing reframed from token-fit check to a three-axis eval: findings quality, cost, reliability; eval-driven rotation discipline named explicitly in the closing paragraph); Catch 35 (§3.5 expanded with the 2026-04-28 race-condition story, \$0.73 synthesis loss, recovery via rebase-before-push commit 05f8c74 and retry-with-backoff commit b5dd4e0; “Cross-vendor heterogeneity guards against vendor-specific model behaviors; race conditions and transient API outages need their own architectural treatment”); Catch 48 (Figure 1 regenerated per Codex: title “Vibe-science (origination)” → “Interactive hypothesis development”; box content dropped “via Claude Code” and “comp-NNN” Open-Enzyme-internal shorthand → “interactive synthesis; picks threads, names sub-claims, queues computation before wet-lab”; more accessible to bioRxiv readers without OE-internal context)
§4 Heterogeneity rationale	Anthropic Claude Opus 4.7	DeepSeek V4-Pro (via OpenRouter) + audit-driven additions	Catch 2 (four-pass framing didn’t match operational daemon, self-caught session 4); Catch 11 (3 overconfident absolutes softened per DeepSeek catches 1–3); Catch 11 cont. (architecture-motivation tone softened per DeepSeek catch 5); Catch 36 (new subsection “Capability profile and exposed subjectivity” added: vendors differ in capabilities (Anthropic prompt caching 91% hit rate, OpenAI structured outputs, Gemini multilingual fluency) not just biases; cross-vendor review exposes subjective verdict-criterion disagreements (Opus 2/3/1 NOVEL/EXT/REST vs GPT-5.5-tuned 0/2/4) rather than resolving them)
§5 Case studies	Anthropic Claude Opus 4.7	DeepSeek V4-Pro (via OpenRouter) on the initial pass; Codex hostile-persona reviews drove the §5.0 leading-example restructure (first round) and the Appendix C catch summary table (second round)	Catch 3 (§5.1 manual-walkthrough understated, self-caught); Catch 4 (§5.3 Km/specific-activity mix-up, self-caught); Catch 5 (§5.4 pre-daemon architecture misdescribed, self-caught); Catch 9 (§5.3 Km factor was 7,500× → corrected to 7.5×; arithmetic error originated in wiki primary source paperclip-deep-dive.md; surfaced by DeepSeek cross-vendor review, landmark reflexive catch) ; Catch 11 cont. (§5.4 “cheaper architectures cannot catch it” softened, DeepSeek catch 6); Catch 41 (Codex first-round: §5.0 added as leading reflexive example pulling the Paperclip 7.5× self-correction up from §5.3’s parenthetical; biology-substrate bracket paragraph added at §5 head); Catch 42 (Appendix C catch summary table added per Codex second-round concrete suggestion: per-row substrate / models / claim / failure class / verification source / downstream consequence)

Section	Primary drafter	Independent reviewer	Catches logged
§5.5 Search-amplifier case (generative class)	Anthropic Claude Opus 4.7 (drafted from logs/v4-peer-review-2026-04-25-deepseek + logs/v4-peer-review-2026-04-25-gemini + operations/notable-moments.md 2026-05-08 entry as primary sources)	Codex hostile-persona reviews engaged this case alongside the §5 ; no specific Codex catch surfaced on the §5.5 prose itself, but second-pass review explicitly approved the search-amplifier framing as part of the “no longer easy to dismiss” verdict	Anchored on the 2026-04-25 double-peer-review convergence (DeepSeek + Gemini independently flagged the androgen-urate axis, carnosine-as-androgen-counter-agent, and fructose-challenge findings as “Missed by Claude”). Downstream impact verified against the project’s actioned record: three numbered computational experiments (016, 017, 019), one hypothesis card (H-AN-08), one new wiki page (androgen-natural-modulation.md). Catch 37 (new failure class added to §5: contamination distributing identically across vendors via training data, defended by V-of-V chains targeting primary source rather than by cross-vendor heterogeneity; three-layer worked example with the eurycomanone GOUT-UNFAVORABLE → GOUT-FAVORABLE direction-of-effect reversal as the consequential layer)
§5.6 Corpus-level contamination / V-of-V (added Session 9)	Anthropic Claude Opus 4.7 (drafted from operations/notable-moments.md 2026-05-07 entry + commits a61f0d9 , c32a623)	Author selection from audit memo (audit-2026-05-13-catch-history.md §1.7 + §3.2); Codex hostile-persona reviews engaged §5.6 across all rounds and the second-pass review explicitly commended it as one of the two strongest additions (“the paper now admits that cross-vendor review is not magic... that concession makes the rest of the argument more credible”)	Catch 14 (Table 1 high-end arithmetic \$1.50 → \$1.30; sum-of-components check by Gemini); Catch 38 (new §6.2 subsection added: verification-discipline cost economics for brief-hygiene re-runs and V-of-V chains, \$5 and \$3-4 per incident respectively); Catch 43 (empirical pilot subsection added per Codex second-pass Class 3.2 ask: 8 prompts × 4 vendors, mean pairwise agreement 82.3%); Catch 46 (within-vendor replicate study added: partition of 17.7% cross-vendor disagreement into 9.2% within-vendor noise + 8.5% real cross-vendor heterogeneity; reasoning-mode-stability hypothesis falsified); Catch 47 (cross-vendor manuscript audit-trail catch documented in §6 meta-catch box: Codex surfaced that the manuscript prose, the claim-coding file, and the raw gemini-r1.md response artifact disagreed on Gemini’s r1 PTS1 answer; mismatch survived two prior hostile-persona review passes that focused on prose and architecture rather than artifact-level reconciliation; corrected claim file preserves the original wrong coding as audit_correction field so the trail stays inspectable) ; Catch 50 (variance-partition language softened per third-pass Codex nit: “approximately,” “under the pilot’s manual claim-coding scheme”)
§6 Operational data + Empirical pilot	Anthropic Claude Opus 4.7 (direct log extraction for cost / latency; empirical-pilot subsection drafted by sub-agent off the empirical-vendor-comparison/artifacts); within-vendor replicate study by a separate sub-agent	Google Gemini 2.5 Pro (via OpenRouter) on the original §6 prose; Codex final-check pass (2026-05-14) caught the PTS1 / Gemini-r1 / claim-coding audit-trail mismatch that produced the §6 meta-catch box and surfaced figure-2 clipping that drove the regeneration	
§7 Limitations	Anthropic Claude Opus 4.7	Google Gemini 2.5 Pro (via OpenRouter) + author read-pass; Codex hostile-persona reviews engaged §7 across all four rounds	Catch 15 (prompt-brittleness limitation added per Gemini, framing later replaced); Catch 16 (“data-heavy no value” softened per Gemini); Catch 34 (Catch 15’s hypothetical “silent vendor update” framing replaced with the actual operational record: the GPT-5.5 launch + initial too-narrow eval + prompt retune per OpenAI’s published guidance, which flipped Pass 3 results to 5× cheaper); Catch 44 cont. (“strictly stronger” softening touched §7 limitations framing)

Section	Primary drafter	Independent reviewer	Catches logged
§8 Discussion	Anthropic Claude Opus 4.7	Author read-pass; Codex hostile-persona reviews engaged §8 across all four rounds (rhetorical-force calibration on the discussion was driven by Codex’s “tone the variance-partition language down one notch” feedback)	Catch 39 (§8 reflexive closing tightened to leverage the audit trail directly: named the cross-vendor review distribution, the reviewer-surfaced / author-surfaced split, and four representative catches that map onto §5 failure classes); Catch 41 cont. (audit-trail-total count updated from 34 catches to ~50 catches reflecting the additional Codex-review rounds + replicate study + meta-catch + final cleanup catches)
§9 Conclusion	Anthropic Claude Opus 4.7	Codex hostile-persona reviews (final-check verdict: “post it. This has enough novelty, enough auditability, and enough self-inflicted evidence to be worth being in the world.”)	Catch 41 cont. (§9 “five representative catches” updated to “seven case studies plus pilot empirical study” reflecting the §5.0 + §6 additions); Catch 49 cont. (“literature” dropped from the conclusion’s framing alongside the title-and-abstract change)
Methods Appendix	Anthropic Claude Opus 4.7 (context unique to drafter)	Codex final-check pass reviewed the Methods Appendix narrative; the appendix’s catch-count and review-pass numbers were updated to reflect the additional Codex rounds	Catch 1 (primary drafter confabulated PaperOrchestra’s existence on first ask, caught by user push-back). Preserved verbatim in the Methods Appendix as central reflexive demonstration. Subsequent updates: catch-count totals revised from 34 to ~50 to reflect the Codex review rounds; cross-vendor-review-distribution prose updated to credit Codex’s role across the post-initial-drafting cycles.
Figure 1	Anthropic Claude Opus 4.7 (matplotlib script)	Author read-pass (multiple); Codex final-check pass regenerated the figure	Catch 33 (overlapping arrows, text outside boxes; redrawn with vertical-spine alignment for cycle elements, explicit-segment loop with perpendicular arrival at Author/Opus, right-angle elbow on the peer-review branch, Layer 1 title moved to give the Author box breathing room, substrate arrow re-anchored to originate from Layer 1 rather than from the wiki corpus); Catch 48 (Codex regenerated 2026-05-14: Layer 1 title “Vibe-science (origination)” → “Interactive hypothesis development”; box content dropped “via Claude Code” + “comp-NNN” Open-Enzyme-internal shorthand → “interactive synthesis; picks threads, names sub-claims, queues computation before wet-lab”)
Figure 2	Anthropic Claude Opus 4.7 (matplotlib script)	Author read-pass; Codex external review (round 1 surfaced the taxonomy reframing; final-check pass regenerated layout to fix clipping)	Catch 24 (recast as qualitative taxonomy per Codex external-review feedback); Catch 40 (Session 9 redraw to include §5.6 as a fifth cell with a new “verification-of-verification” surfacing color, layout widened from 1×4 to 1×5); Catch 48 cont. (Codex regenerated 2026-05-14 to fix surfacing-band clipping: widened figsize 16×6.2 → 18×7.0, cell dimensions proportionally enlarged, surfacing-band height 0.28 → 0.32, font reduced 8.5 → 7.3, three surfacing labels re-wrapped, V-of-V chain abbreviation tightened)
Table 1	Anthropic Claude Opus 4.7 (direct log extraction)	Gemini 2.5 Pro (arithmetic check)	Catch 14

Cross-vendor review run summary: four parallel review jobs were orchestrated via OpenRouter (DeepSeek V4-Pro on §4+§5, Gemini 2.5 Pro on §3+§6+§7, Claude Opus 4.5 and DeepSeek V4-Pro in parallel on §2). Total OpenRouter cost ~\$0.18; total wall time ~131 seconds (parallel execution). 20 catches surfaced across the four

reviewers: 12 substantive corrections applied to the manuscript, 1 correction propagated to the underlying wiki primary source, 2 reviewer false positives logged for transparency. Verbatim model outputs preserved as supplementary material.

Subsequent review passes. After the initial cross-vendor pass, the manuscript was reviewed by Codex (a commercial coding/research agent), which surfaced six recommendations (stale metadata, “proof” vs “case-study” framing, missing adjacent-literature citations, Figure 2 recast as taxonomy not frequency distribution, Zenodo DOI as a submission gate, reference-scheme harmonization). All six applied as Catches 21–26. The author then ran a wide-scope audit of the project’s git history and operational record for catch moments not yet in §5 (`audit-2026-05-13-catch-history.md`), which informed subsequent §3/§6/§7 revisions. A final author read-pass surfaced eight additional catches (27–34), including a self-caught Pass 3 model-name error, internal-routing rationale correction, figure layout corrections, and the §7 prompt-brittleness reframing described in Catch 34. **Two further substantial revision rounds** followed: an external hostile-persona review pass (Catches 35–45 across two cycles: Class 1 mechanical sweeps, Class 2 biology-bracket + Appendix C catch summary table, Class 3.2 §6 empirical pilot + literature grounding, Class 4.7 §5.0 leading reflexive example, and a second-pass reference-cleanup + rhetorical-softening round); and a within-vendor replicate study (Catch 46) addressing the second-pass reviewer’s pilot-strength concern by partitioning the original cross-vendor disagreement into temperature noise versus real heterogeneity. A final cross-vendor manuscript audit (Catch 47) detected the PTS1 / Gemini-r1 claim-coding mismatch documented in the §6 meta-catch box: the manuscript prose, the claim-coding file, and the raw response artifact were three layers that disagreed with each other, and the disagreement surfaced only when a different model family was asked to verify across the three layers. Final figure and scope cleanup added Catches 48–50 (figure regeneration, title/scope tightening, and post-audit number harmonization). **Audit-trail total at submission: approximately 50 catches** logged across the revision history, of which roughly 20 are author-or-self-caught and roughly 30 are surfaced by independent reviewers (cross-vendor manuscript passes plus external hostile-persona review plus the cross-vendor audit-trail reconciliation and final-check pass). The per-catch detail and discoverer attribution are in `revisions.md`; the totals above are approximate because catch boundaries are sometimes blurry (a catch that surfaces during one pass and is closed during a follow-up pass is attributed to the surfacing pass).

Appendix B, Revision log (*see `revisions.md` for the working log*)

Appendix C, Catch summary table

The case studies in §5 are summarized below. Each row anchors a single catch to its substrate, the model(s) involved, the original claim or gap, the class of failure, what surfaced the catch, and the downstream consequence in the corpus. §5.0 is the leading reflexive example, §5.1–§5.6 are operational catches from the 2026-04-25 to 2026-05-08 window, and the final row documents the §6 meta-catch (a second reflexive case study where the pilot’s own claim-coding mismatched the raw response artifact and was caught by an independent cross-vendor audit pass). This table is intended for readers who want the evidence at a glance before engaging the case-study prose.

Case	Substrate	Model(s) involved	Original claim or gap	Failure class	Verification source	Downstream consequence
§5.0 Paper catches an error in its own source material (2026-05-13)	This manuscript’s draft §5.3 prose, citing <code>wiki/paperclip-deep</code> documentation of a Paperclip MCP misreport	Anthropic Claude Opus 4.7 (drafter of both the wiki page and the §5.3 prose); DeepSeek V4-Pro (cross-vendor reviewer that caught the magnitude error during manuscript drafting)	Wiki page claimed Paperclip MCP misreported Najjari 2022 Km by “~7,500×”; the correct magnitude is $\sim 7.5 \times (52.61 \mu\text{M})$ true value vs 7 μM reported)	Recursive within-vendor cascade: the wiki page documenting a tool hallucination itself contained a hallucinated magnitude; the within-vendor pipeline that produced both the wiki and the §5.3 draft did not surface either error	Cross-vendor review of the present manuscript on 2026-05-13, with explicit primary-source-cross-check brief; the reviewer compared draft prose against the original Najjari 2022 paper (PMC9773812)	Both the wiki page and §5.3 corrected (commit <code>bcf6bae</code>); the catch is verifiable as a literal diff in the manuscript’s git history; preserved as Appendix B Catch 23

Case	Substrate	Model(s) involved	Original claim or gap	Failure class	Verification source	Downstream consequence
§5.1 DAF SCR1-4 disulfide-count (2026-05-06)	comp-012 stalk-truncation analysis of human DAF/CD55 (UniProt P08174), wiki page daf-cd55-scr14-truncation	Anthropic Claude Sonnet (subagent that authored the page); pipeline-internal sweep and manual architectural surfaced the catch	“3 conserved disulfide bonds per SCR domain → 12 total for SCR1-4” asserted in four places of prose, despite the comp-012 pipeline’s stated “Disulfide bonds not modelled” limitation	Within-vendor cascade: single-model hallucination propagated overnight to the H05 hypothesis card and downstream chaperone-load calculation (29-disulfide panic)	UniProt P08174 primary record (8 DISULFID annotations, 2 per SCR domain, canonical sushi/CCP fold)	Pre-commit grep-verify gate added to wiki authoring discipline (CLAUDE.md Rule 4); correction note logged on comp-012 page
§5.2 comp-018 framing bias (2026-05-08)	Brief for upstream complement modulator literature scan; user phrasing “if it’s rosemary I’ll grow rosemary” landed verbatim	Anthropic Claude (comp-018 subagent that surfaced rosmarinic acid as Tier-1); independent rerun (comp-020) caught the contamination	Rosmarinic acid as headline Tier-1 dietary CP0 modulator, with thinner evidence than the report narrative implied	Upstream-of-subagent contamination: user-supplied framing biased the subagent’s prioritization; invisible to within-pass review because output was internally coherent	comp-020 verification rerun under a deliberately brief-scrubbed prompt; rhetorical-callback tell (“Brian’s literal framing landed empirically”)	Subagent brief hygiene discipline codified (scripts/SWEEP-ARCHITECTURE.md); rerun surfaced three tied Tier-1 candidates (rosmarinic acid, <i>Helicteres benzofuran</i> lignans, luteolin) and revealed missed coverage
§5.3 Paperclip MCP probe (2026-05-05)	~12 uricase-variant papers with hand-curated wiki ground truths for organism, kinetic parameters, modality	Paperclip MCP map operator (external tool, vendor-independent)	Structured extraction across the 12 papers, evaluated for pipeline integration	External-tool reliability: (a) wrong organism IDs (<i>A. flavus</i> → <i>A. globiformis</i> ; porcine-human chimera → <i>P. chrysogenum</i>); (b) 7.5× Km misreport for Najjari 2022 (PMC9773812); (c) invented wet-lab data for purely-computational papers	Ground-truth probe against curated wiki facts before pipeline integration	Decision NOT to integrate Paperclip into Pass 1 or Pass 2; remains a manual research tool requiring grep-verify on every output
§5.4 DeepSeek Connection 7 (2026-04-25)	Open Enzyme wiki at commit 4a40f74 , sole-Claude-driven prior to that date	DeepSeek V4-Pro (peer-review pass) named the gap; Anthropic Claude Opus 4.7 (the dominant pipeline that had missed it)	Methodological observation: single-model synthesis converges on that model’s blind spots and biases; the prior pipeline had not surfaced this risk	Methodology gap invisible from within the dominant vendor’s pipeline; the highest-value catch class because within-vendor review is structurally unlikely to surface it	Independent cross-vendor peer-review pass on identical substrate	Formalization of the three-pass cross-vendor daemon architecture; the seminal catch that motivated this paper’s approach
§5.5 Double-peer-review convergent generative catches (2026-04-25)	Same Claude Opus 4.7 substrate at commit 4a40f74	DeepSeek V4-Pro and Google Gemini 2.5 Pro running independent peer-review passes; both converged on three findings the substrate had not made	Three structural connections: (1) androgen-urate axis as therapeutic ceiling, not stratification variable; (2) carnosine as androgen-specific URAT1 counter-agent; (3) fructose challenge as acute n=1 efficacy readout	Structural under-connection of corpus content (generative, not defensive); within-vendor pipeline read the source pages and didn’t make the connections	Independent cross-vendor double-peer-review with convergence between two distinct-prior vendors	3 new numbered computational experiments (comp-016, comp-017, comp-019); 1 hypothesis card (H-AN-08); new wiki page androgen-natural-modulation.md ; fructose challenge protocol added to validation-experiments.md and self-experiment-protocol.md . Total cost \$0.87 for the two peer-review passes

Case	Substrate	Model(s) involved	Original claim or gap	Failure class	Verification source	Downstream consequence
§5.6 Eurycoma / tongkat ali corpus-level contamination (2026-05-07)	Testosterone-axis adjuvant literature, three-layer verification chain on cited evidence	Multilingual literature-scan subagent (within-pipeline) propagated the contaminated claims; comp-015 v2 verification chain surfaced them	(1) “37% testosterone elevation” from Talbott 2013 (salivary, mixed-sex, moderately-stressed, not serum free T in hypogonadal men); (2) “Shin KH 2024 enclomiphene” citation that doesn’t exist in PubMed (real citation: Saffati 2024 PMID 39434750); (3) eurycomanone tagged GOUT-UNFAVORABLE based on a claimed-but-unsupported XO-inhibitor mechanism	Corpus-level contamination from training data: the only failure class in the taxonomy where cross-vendor heterogeneity provides no defense because all vendors regurgitate the laundered claim	Verification-of-verification chain: independent subagent reading the cited PMIDs’ primary text directly (same vendor as original synthesizer; the discipline, not the vendor diversity, was load-bearing)	Wiki tag flipped GOUT-UNFAVORABLE → GOUT-FAVORABLE for eurycomanone (commit c32a623); Saffati 2024 citation correction; V-of-V chain pattern codified as defense for this class. Cost ~\$3-4 in subagent tokens
§6 meta-catch PTS1 claim-coding audit-trail mismatch (2026-05-14)	This manuscript’s §6 pilot prose + claim-coding file 01-uricase-mechanism-factual.json C1.5 + raw response artifact gemini-r1.md (three layers that disagreed with each other)	Anthropic Claude Opus 4.7 (subagent that authored the 01-uricase-mechanism-factual.json coding); Anthropic Claude Opus 4.7 (orchestrator that integrated §6 into the manuscript); Codex (OpenAI-family, independent model that surfaced the mismatch)	Claim-coding file recorded Gemini’s r1 PTS1 answer as “disagree / SSH (Ser-Ser-His)”; manuscript prose inherited the wrong coding; raw response artifact gemini-r1.md line 30 contained the canonical “SKL (Ser-Lys-Leu)” affirmation. The mismatch passed through two within-vendor revision cycles (initial drafting plus two hostile-persona review passes)	Recursive same-vendor cascade across three layers of the audit trail: the empirical pilot is the case where the §5.1 within-vendor-cascade failure mode bit the paper’s own empirical evidence	Independent cross-vendor manuscript audit (Codex, OpenAI-family), comparing draft prose against the raw response files rather than against the intermediate claim-coding file	Claim-coding file corrected (original wrong coding preserved as <code>audit_correction</code> field so the trail stays inspectable); cross-vendor and within-vendor agreement matrices re-run on corrected data; manuscript prose updated in §6 with the new numbers (cross-vendor 82.3% rather than 81.8%, factual recall 70.0% rather than 63.3%, Gemini-vs-OpenAI 79.6% rather than 77.1%); Finding-1 narrative rewritten as 2-of-3-correct rather than 3-of-3-different; meta-catch box added to §6 to keep the scar visible in the audit trail

Data and Code Availability

All Open Enzyme research-wiki content, sweep-daemon code and prompts, operational logs, and the drafting workspace for this manuscript are publicly available at <https://github.com/brianpabent/open-enzyme>. The submission-time snapshot of the repository, including the full empirical-vendor-comparison pilot artifacts (raw vendor responses, claim codings, and the reproducible analysis pipeline), is archived on Zenodo as the “**Epistemic Heterodoxy**” release: version-specific DOI [10.5281/zenodo.20190078](https://doi.org/10.5281/zenodo.20190078) (the immutable archive of the exact state cited in this manuscript), concept DOI [10.5281/zenodo.20190077](https://doi.org/10.5281/zenodo.20190077) (resolves to the latest version of the release as the corpus evolves). Source repository tag: `epistemic-heterodoxy-v1`. Specific artifacts referenced inline in this manuscript:

- **Sweep daemon engineering doc:** `scripts/SWEEP-ARCHITECTURE.md` (architecture, failure modes, operational hardening)
- **GitHub Actions workflow:** `.github/workflows/wiki-sweep.yml` (the daemon’s CI orchestration)

- **Pass prompts:** `scripts/sweep-prompt-1-propagate.md`, `scripts/sweep-prompt-2-synthesize.md`, `scripts/sweep-prompt-3-review.md`
- **Platform thesis with heterogeneity-guard rationale:** `wiki/open-source-platform.md` §“Multi-model synthesis as guard against epistemic homogenization”
- **Working instructions / pre-commit grep-verify gate:** `CLAUDE.md` §“Pre-commit grep-verify gate for load-bearing numbers”
- **Case-study primary sources:** `wiki/daf-cd55-scr14-truncated-computational.md` §1.5 (§5.1); `operations/comp-018-vs-comp-020-retrospective.md` (§5.2); `wiki/paperclip-deep-dive.md` (§5.3); `logs/v4-peer-review-2026-04-25-deepseek.md` (§5.4)
- **Empirical pilot artifacts (§6):** `papers/cross-vendor-heterogeneity-guard/empirical-vendor-comparison/` (prompts/, responses/ with SHA-256-anchored raw vendor outputs across r1/r2/r3 replicates, outputs/claims/ atomic-claim verdict files including the §6 meta-catch audit_correction trail, `run_experiment.py` and `analysis.py` reproducible pipeline, `analysis_within_vendor.py` for the partition study, `provenance.md` per-claim verification log, `_cost_log.jsonl` for per-call cost reconstruction).
- **Literature-scan artifacts (§6):** `papers/cross-vendor-heterogeneity-guard/literature-scan-inter-vendor-agre` (annotated bibliography, key-quotes file, synthesis, provenance + search log).
- **Manuscript drafting workspace:** `papers/cross-vendor-heterogeneity-guard/` (`draft.md`, `revisions.md` = Appendix B raw log, `paperorchestra-workspace/` for §2 production records including verbatim cross-vendor review outputs at `reviews/`, `figures/` for matplotlib sources)
- **Author’s contextual analysis of the AI-scientist landscape:** Abent B. “[Grounding the AI-Scientist Hype](#)” personal blog, 2026-05-08.

References

1. Du Y, Li S, Torralba A, Tenenbaum JB, Mordatch I. *Improving Factuality and Reasoning in Language Models through Multiagent Debate*. ICML 2024. arXiv:2305.14325. 2023.
2. Madaan A, Tandon N, Gupta P, Hallinan S, Gao L, Wiegrefe S, et al. *Self-Refine: Iterative Refinement with Self-Feedback*. NeurIPS 2023. arXiv:2303.17651. 2023.
3. Shinn N, Cassano F, Labash B, Gopinath A, Narasimhan K, Yao S. *Reflexion: Language Agents with Verbal Reinforcement Learning*. NeurIPS 2023. arXiv:2303.11366. 2023.
4. Verga P, Hofstätter S, Althammer S, Su Y, Piktus A, Arkhangorodsky A, et al. *Replacing Judges with Juries: Evaluating LLM Generations with a Panel of Diverse Models*. arXiv:2404.18796. 2024.
5. Ouyang L, Wu J, Jiang X, Almeida D, Wainwright C, Mishkin P, et al. *Training Language Models to Follow Instructions with Human Feedback*. NeurIPS 2022. arXiv:2203.02155. 2022.
6. Bai Y, Kadavath S, Kundu S, Askell A, Kernion J, Jones A, et al. *Constitutional AI: Harmlessness from AI Feedback*. arXiv:2212.08073. 2022.
7. Lee H, Phatale S, Mansoor H, Lu K, Mesnard T, Bishop C, et al. *RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback*. ICML 2024. arXiv:2309.00267. 2023.
8. Lu C, Lu C, Lange R, Foerster J, Clune J, Ha D. *The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery*. arXiv:2408.06292. 2024.
9. Yamada Y, Lange R, Lu C, Hu S, Lu C, Foerster J, et al. *The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search*. arXiv:2504.08066. 2025.
10. Song Y, Song Y, Pfister T, Yoon J. *PaperOrchestra: A Multi-Agent Framework for Automated AI Research Paper Writing*. arXiv:2604.05018. 2026.
11. Shumailov I, Shumaylov Z, Zhao Y, Papernot N, Anderson R, Gal Y. *AI models collapse when trained on recursively generated data*. Nature, 2024. doi:10.1038/s41586-024-07566-y.
12. Shumailov I, Shumaylov Z, Zhao Y, Gal Y, Papernot N, Anderson R. *The Curse of Recursion: Training on Generated Data Makes Models Forget*. arXiv:2305.17493. 2023.
13. Peterson AJ. *AI and the Problem of Knowledge Collapse*. AI & Society. arXiv:2404.03502. 2024.
14. Wright D, Masud S, Moore J, Yadav S, Antoniak M, Christensen PE, et al. *Epistemic Diversity and Knowledge Collapse in Large Language Models*. arXiv:2510.04226. 2025.
15. Rudko I, Bashirpour Bonab A. *ChatGPT is incredible (at being average)*. Ethics and Information Technology. doi:10.1007/s10676-025-09845-2. 2025.
16. Go J, et al. *LiRA: A Multi-Agent Framework for Reliable and Readable Literature Review Generation*. arXiv:2510.05138. 2025.

17. Kim E, et al. *Correlated Errors in Large Language Models*. arXiv:2506.07962. 2025. (Frontier-convergence finding: 350+ LLMs surveyed; co-error rate ~60% on a shared leaderboard task; model scale negatively correlates with epistemic diversity.)
18. Yuan H, et al. *Do Mixed-Vendor Multi-Agent LLMs Improve Clinical Diagnosis?* arXiv:2603.04421. 2026. (Mixed-vendor team o4-mini + Gemini-2.5-Pro + Claude-4.5-Sonnet hit +10pp absolute recall@1 over best single-vendor model on the hardest 40-case rare-disease subset; lift inversely correlates with cross-vendor Jaccard similarity, 0.269–0.750.)
19. Amiri-Margavi A, et al. *Enhancing Answer Reliability Through Inter-Model Consensus of Large Language Models*. arXiv:2411.16797. 2025. (N=100 PhD-level statistics questions; Fleiss kappa across GPT-4 / Claude-3-Opus / Gemini-1.5-Flash / LLaMA-3-70B spans 0.7160 to 0.2811; full-agreement rates 65–86% across model-generated question sources.)
20. Zhang Y, et al. *Stop Overvaluing Multi-Agent Debate, We Must Rethink Evaluation and Embrace Model Heterogeneity*. arXiv:2502.08788. 2025. (“Model heterogeneity acts as a universal antidote” to homogenized-model failure in multi-agent-debate methods; homogeneous-model debate often fails to beat single-agent baselines.)